



Technical Proposal

Weather Monitoring and Alerting System

Sydney Metro M1 Line

*In response to Part A — Statement of Requirements (Version 01, 03/07/2026)
Issued by Metro Trains Sydney (MTS)*

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1. Cover letter

22 July 2026

Attention: Contract Management Representative
Metro Trains Sydney (MTS)

Re: Response to Part A — Statement of Requirements — Weather Monitoring for the Sydney Metro M1 Line

Observator Instruments welcomes the opportunity to respond to Metro Trains Sydney's Statement of Requirements for weather monitoring and alerting across the Sydney Metro M1 line. Observator is a specialist manufacturer and systems integrator of meteorological and hydrological monitoring solutions, established in 1924 and ISO 9001:2015 certified, with a local presence in Australia and a global installed base spanning aviation, rail, road, port and water-management infrastructure.

We have read the Statement of Requirements in full and confirm that we understand the safety-critical nature of the Services: rainfall, flood water level, temperature and wind monitoring, together with automatic control of the existing trackside flood pumps at Marrickville, feeding a resilient alerting and reporting platform that supports MTS operational decisions such as front-of-train patrols, temporary speed restrictions and blocking of the line.

The solution described in this proposal is built around Observator's modular Blue2 approach — sensing, data transport, data publishing and lifecycle support — configured specifically for the seven nominated locations. Each station combines a Gill WindSonic 75 anemometer for wind, a Gill GMX300 compact sensor for temperature and humidity (and barometric pressure), a dedicated RIMCO 7499 BOM-standard rain gauge for rainfall, and YGRD-65-D terahertz radar water-level sensing with a backup float switch at flood sites, all wired to an OMC-048 data logger feeding a central server — each station is fitted only with the sensors its location requires. Every station is self-powered by solar and communicates over cellular 4G/5G, in line with the assumption that no mains power or existing communications network is available at site.

Where the Statement of Requirements notes that thresholds, alert wording, kilometrages and GPS positions are "pending confirmation", our platform is fully configurable and these values will be finalised with MTS during project development. A small number of technical points that warrant confirmation against current datasheets, and the specialist pump supply, are listed transparently in Section 15 (Assumptions, exclusions and clarifications) rather than being glossed over.

We would be pleased to attend a site visit and to develop this response into a detailed design. Please direct any queries to the contact below.

Yours faithfully,



YeHuat

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2. Executive summary

Observator proposes a fully managed weather monitoring and alerting system for the Sydney Metro M1 line comprising seven autonomous weather stations, an integrated trackside flood-pump control package at Marrickville, cellular data transmission, a scriptable alerting engine, and a secure online portal — all backed by local maintenance and calibration under our Blue2Care service program.

Why Observator

- **Manufacturer and integrator, single point of accountability.** Observator both manufactures instruments and integrates best-in-class sensors from partners such as Gill and RIMCO, delivering turnkey systems "from a single sensor to a complete measurement station including telemetry and data processing". MTS deals with one accountable party across sensors, telemetry, server and service.
- **Rail and infrastructure heritage.** Observator's infrastructure portfolio already addresses railways, bridges and tunnels, monitoring road condition, visibility, wind, temperature and water level — the exact parameters in this Statement of Requirements.
- **Local support.** An Australian office (Melbourne) provides site attendance, maintenance and calibration, supporting the response-time obligations in the Statement of Requirements.
- **Configurable, safety-critical alerting.** The OMC-048 data logger and the central server implement the multi-stage rainfall, flood, temperature and wind thresholds — including the vigilance-timer logic (6 h / 12 h / 48 h) and automatic redundancy fallback — and deliver screen, email and SMS alerts within the required timeframes.
- **Quality and sustainability.** ISO 9001:2015 certification and the Blue2Green sustainability program underpin delivery.

Solution at a glance

Element	Observator provision
Weather stations	7 autonomous stations at the nominated locations (Marrickville, Marrickville–Dulwich Hill, Canterbury, Campsie, Belmore triangle, Lady Game Drive tunnel, Windsor Road) — sensor mix per Section 5.
Sensors	A Gill WindSonic 75 ultrasonic anemometer (0-75 m/s) for wind, a Gill GMX300 compact temperature and humidity sensor, and a dedicated RIMCO 7499 BOM-standard tipping-bucket rain gauge for rainfall - each fitted only at the locations that require that parameter; YGRD-65-D terahertz radar water-level sensors with a contact float switch (and manual gauge boards) at flood locations.
Flood pumps	Integration to the existing trackside pumps at Marrickville - the OMC-048 logger triggers the existing pumps in real time from the level sensor (float switch as an independent backup) and reads their run/fault status for alerting. Observator integrates to the existing pumps and control gear rather than supplying pumps.
Telemetry	OMC-048 scriptable data logger per station (real-time pump control, 1-minute logging, configurable transmit) with modem, PSU and battery in a pole-mounted cabinet, 4G/5G cellular, autonomous solar power.
Alerting	Configurable threshold engine; screen, email + SMS within 5 min (weather) / 30 min (system faults); automatic redundancy fallback and self-diagnostics.
Portal	Central server — data visualised over a map, laptop/mobile HMI, 10-minute health refresh, alert interrogation/acknowledgement, and historical data storage and retrieval.
Service	Blue2Care preventative and corrective maintenance, calibration certificates, and lifecycle support from the Australian office.

3. Company and capability

Observator Group is a Netherlands-headquartered group established in 1924, operating internationally with offices in Australia, Germany, the Netherlands, Poland, Singapore and the United Kingdom. Observator Instruments — the division responsible for this proposal — specialises in meteorological and hydrological monitoring, offering sensors, purpose-built displays and software systems, delivered as anything from a single instrument to a complete measurement network including telemetry and data processing.

The company is ISO 9001:2015 certified and applies formal quality management across product manufacture and system delivery. Its Blue2Green program drives environmental sustainability across operations.

3.1 The Blue2 delivery framework

Observator structures every solution around four modular building blocks. This proposal maps directly onto them:

Building block	Role in this project
Blue2Scan — sensor technology	The Gill WindSonic 75 anemometers, Gill GMX300 compact temperature/humidity sensors, RIMCO 7499 rain gauges, YGRD-65-D radar water-level sensors and float switches installed at the seven locations.
Blue2Link — data transport & storage	OMC-048 data loggers and cellular telemetry that gather, buffer and transmit data, drive the pumps in real time and run the local alert logic.
Blue2Cast — data publishing	The central server, map view and dashboards that present metrics, health status and alerts to MTS and store history.
Blue2Care — uncompromised support	Project management, commissioning, preventative and corrective maintenance, calibration and lab services, and emergency management.

Representative infrastructure and aviation clients include national meteorological and roads authorities and major airports, reflecting experience with safety-critical, always-on monitoring networks.

4. Solution architecture

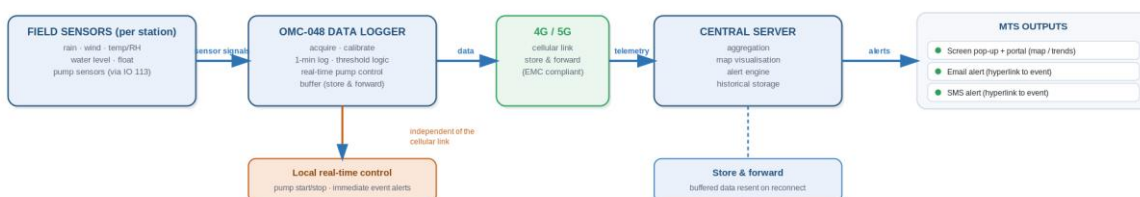
Each of the seven locations is delivered as a self-contained, pole-mounted weather station. Because the Statement of Requirements states that none of the proposed locations have a power source and cannot use the existing rail communications network, every station is designed to be **energy-autonomous and wirelessly connected**:

- A **Gill WindSonic 75 ultrasonic anemometer** at the top of the mast measures wind speed and direction across the full 0-75 m/s (270 km/h) range with no moving parts, meeting the wind-speed range in the Statement of Requirements.
- A **Gill GMX300 compact sensor**, mounted at 1.5-2 m above rail in a shaded position, measures air temperature and relative humidity (and barometric pressure) in a single integrated unit with a built-in radiation screen, outputting digitally to the OMC-048.
- A **dedicated RIMCO 7499 BOM-standard tipping-bucket rain gauge** provides the rainfall measurement at each location where rainfall is a required parameter. It is the single, precision BOM-standard instrument used across the Australian network, and is not fitted where rainfall is not required.
- **At flood locations, a YGRD-65-D terahertz (120 GHz) radar water-level sensor** provides continuous 24×7 non-contact level measurement, complemented by a **contact-point float switch** that both verifies the radar reading and acts as an independent backup trigger for the pumps. The mounting allows flexible installation height to suit each flood plain.
- An **OMC-048 scriptable data logger** in a pole-mounted cabinet (with modem, PSU and battery) acquires every channel, logs 1-minute data, buffers locally, runs the site alert logic, processes the level sensor in real time to trigger the pump, and transmits over 4G/5G cellular (default every 5 minutes, configurable, plus immediate event-triggered transmission on any threshold breach).
- **Autonomous power** is provided by a solar panel and LiFePO4 battery sized for the site load with multi-day autonomy through overcast periods; battery state is itself a monitored, alarmed parameter.
- A **central server** aggregates every station, visualises data over a map, generates and distributes screen/SMS/email alerts, and stores history for retrieval.

4.1 Data and alert flow

Sensors → OMC-048 logger (acquisition, calibration, 1-minute logging, local threshold logic, real-time pump control, buffering) → 4G/5G cellular → central server (aggregation, map visualisation, alert generation, historical storage) → screen, email and SMS to nominated MTS staff, each alert carrying a hyperlink back to the event on the server. Local logic on the OMC-048 ensures that safety-critical detections (for example flood water above rail foot height, or a pump start) are recognised, acted upon and transmitted immediately — independent of the routine telemetry cycle and even under transient loss of central connectivity, with store-and-forward on restoration.

Data and alert flow



Each alert carries a hyperlink back to the event on the server. Safety-critical detections (e.g. water above rail foot, pump start) are recognised, acted on and transmitted immediately - independent of the routine telemetry cycle and even under transient loss of central connectivity.

Data and alert flow (block diagram): field sensors -> OMC-048 -> 4G/5G -> central server -> screen / email / SMS, with local real-time pump control and store-and-forward that continue independently of the cellular link.

4.2 System diagrams

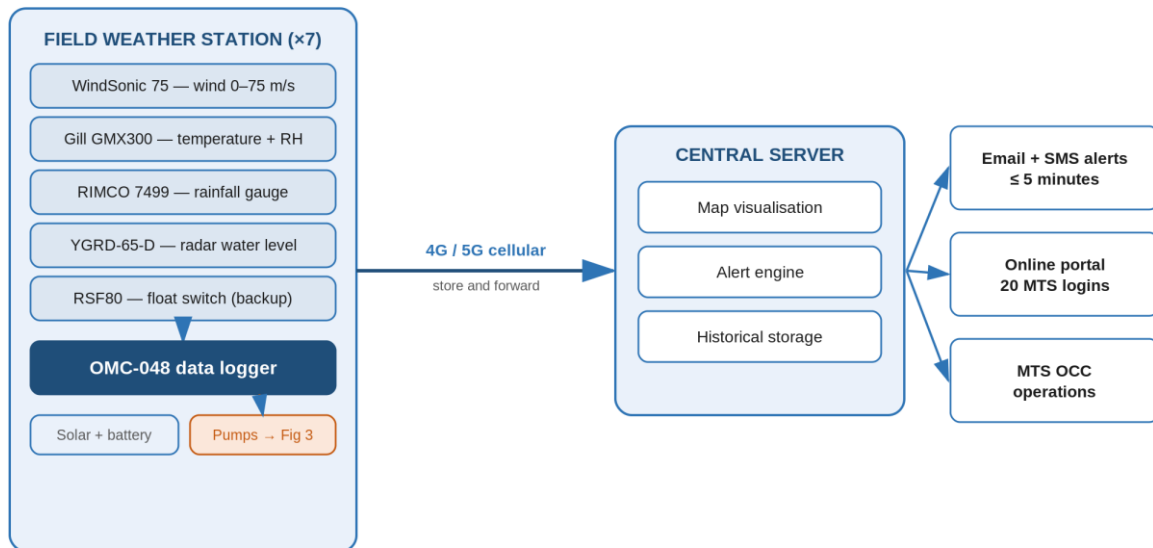


Figure 1 — End-to-end system architecture: seven field stations, cellular telemetry, the central server and MTS alert/portal outputs.

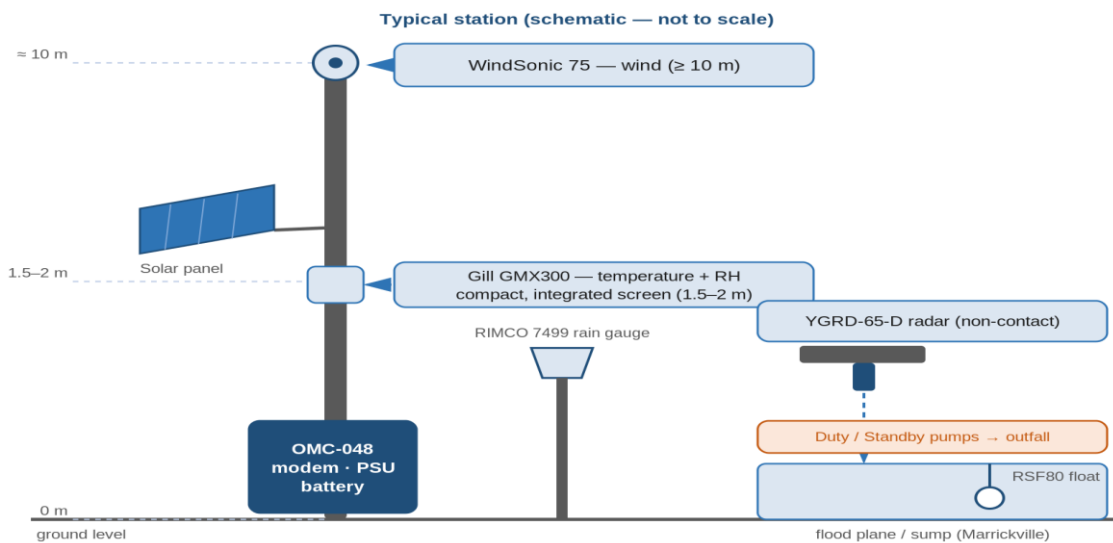


Figure 2 — Typical station layout (schematic, not to scale): mast-mounted sensors with height references, autonomous solar power, and the Marrickville flood site.

Station wiring and OMC-048 connectivity

Typical fully-populated station (e.g. Marrickville / Belmore) - sensor mix varies by location (Section 6)

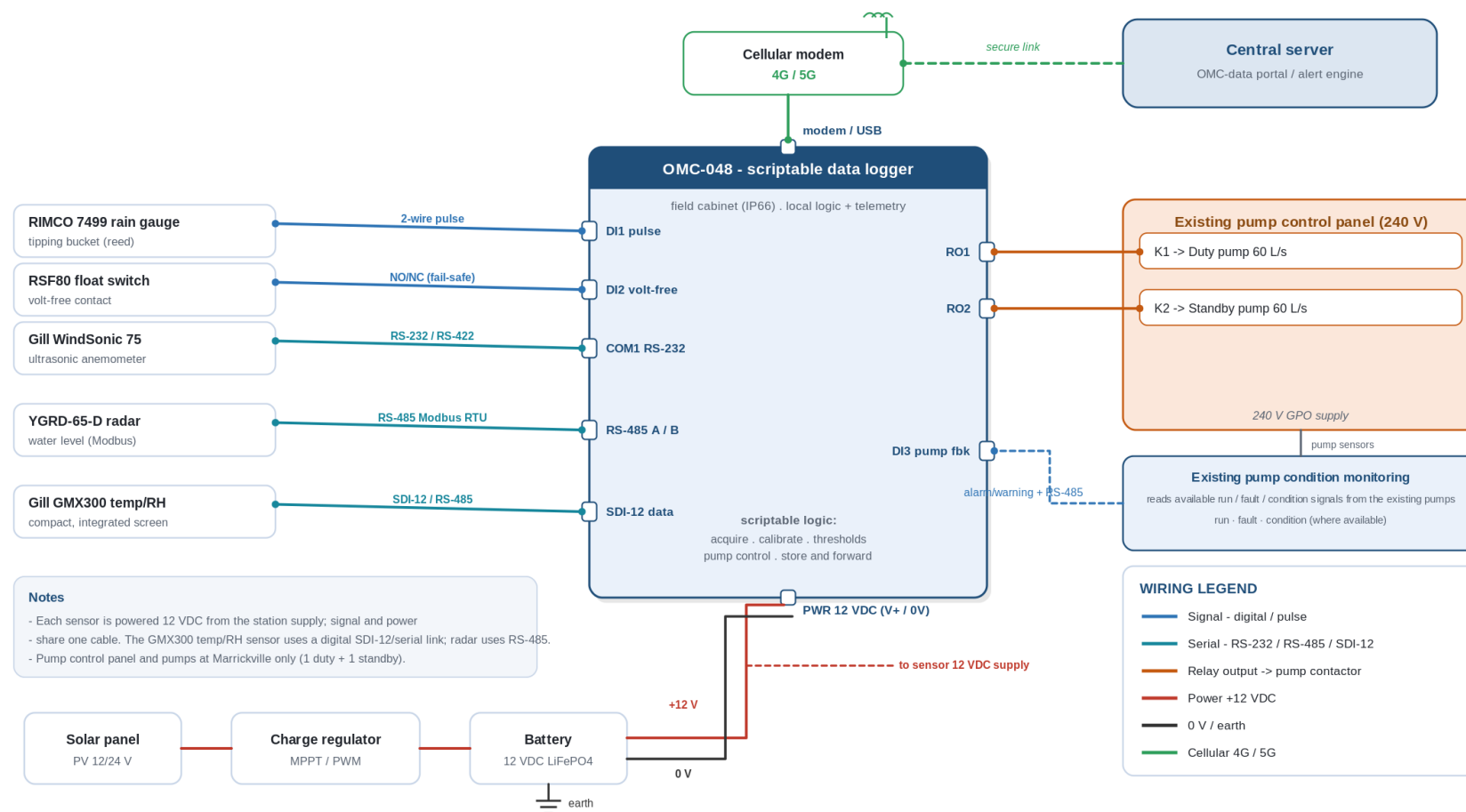


Figure 3 — Station wiring and OMC-048 connectivity: each sensor and its interface wired to the logger (pulse, volt-free, RS-232, RS-485 Modbus, SDI-12), relay outputs to the pump control panel, autonomous solar/battery power, and the cellular link to the central server (fully-populated station; sensor mix varies by location).

4.3 Terminal assignment and per-location variants

The wiring schedule below lists every OMC-048 terminal, the device it serves, its electrical interface and its cabling. Because the sensor mix varies by location (Section 6), the second table shows which devices are fitted at each of the seven stations.

OMC-048 terminal assignment (typical fully-populated station)

Terminal	Device	Interface	Cable	Signals / cores
DI1	RIMCO 7499 rain gauge	Digital pulse (reed switch)	2-core screened	tip pulse, 0 V
DI2	RSF80 float switch	Volt-free contact (NO/NC)	2-core screened	contact, 0 V (fail-safe)
COM1	Gill WindSonic 75	RS-232 / RS-422 serial	4-core screened	Tx, Rx, 0 V, +12 V
RS-485 A/B	YGRD-65-D radar level	RS-485 Modbus RTU	2-pair screened	A, B, 0 V, +12 V
SDI-12 / COM2	Gill GMX300 temp/RH (+ pressure)	Digital (SDI-12 / RS-232 / RS-485)	3-4 core screened	data, 0 V, +12 V
DI3	Pump control panel	Volt-free status	2-core screened	run / trip, 0 V
RO1 / RO2	Pump panel — Duty / Standby	Relay (volt-free) to contactor coil	2-core each	NO, COM
PWR (V+ / 0V)	Battery 12 VDC (solar-charged)	DC power input	2-core	+12 V, 0 V
Modem / USB	Cellular modem (4G / 5G)	USB / serial	—	data, power
Earth	Cabinet / mast bond	Protective earth	earth conductor	PE

Per-location wiring variants

Location	Rain gauge (DI1)	Radar level (RS-485)	Float switch (DI2)	GMX300 T/RH (SDI-12)	WindSonic (COM1)	Pumps (RO1-2)
1 Marrickville	✓	✓	✓	—	—	✓
2 Marrickville–Dulwich Hill	—	✓	✓	✓	✓	—
3 Canterbury	—	✓	✓	—	—	—
4 Campsie	—	✓	✓	—	—	—
5 Belmore triangle	✓	✓	✓	✓	✓	—
6 Lady Game Drive (tunnel)	—	✓ ×2	✓ ×2	—	—	—
7 Windsor Road SSC	—	—	—	✓	✓	—

✓ = fitted; — = not required at that location. Location 6 (Lady Game Drive tunnel) carries two radar level sensors and two float switches (up and down tunnels). The 240 V pump panel and pumps are at Marrickville (Location 1) only. The Gill GMX300 compact temperature/humidity sensor is fitted wherever temperature/humidity monitoring is required; the RIMCO 7499 gauge is the rainfall instrument at the rain locations only.

5. Sensor-by-sensor technical solution

The tables below restate each requirement from Section 3.3.2 of the Statement of Requirements and Observer's proposed provision. Where a value is marked **(confirm)**, the parameter will be verified against the current product datasheet and factory calibration certificate during detailed design; where marked **(configurable)**, it is a software setting finalised with MTS.

5.1 Rainfall measurement

Proposed products: a **RIMCO 7499 BOM-standard tipping-bucket rain gauge** as the sole precision rainfall instrument at each location where rainfall is a required parameter. It is the tipping-bucket gauge used across the Australian Bureau of Meteorology network, giving well-understood, locally proven performance, and it is not fitted where rainfall is not required.



RIMCO 7499 BOM-standard tipping-bucket rain gauge (manufacturer photo).

Requirement	Solution response
Sensor type — tipping bucket or approved equivalent	Comply. RIMCO 7499 tipping-bucket gauge, the BOM-standard instrument.
Measurement range 0–200 mm/hr (min)	Comply. Gauge operates across and beyond 0–200 mm/hr.
Accuracy $\pm 2\%$ ≤ 50 mm/hr; $\pm 3\%$ > 50 mm/hr	Comply (confirm). The RIMCO 7499 is the BOM-standard precision gauge; the required tolerance is verified by factory calibration certificate.
Resolution 0.2 mm per tip (or better)	Comply. 0.2 mm tip (0.1 mm option).
Reporting — real-time event-based + 10-min rolling totals	Comply (configurable). Event-based tip logging plus configurable rolling totals (10-min, 1/6/24-hour windows) in the OMC-048 logger.
Ingress protection IP66 (min)	Comply (confirm). Rugged corrosion-resistant construction; IP rating confirmed on the selected model.
Self-cleaning / anti-blockage; avoid splash locations	Comply. Bird-guard / anti-blockage options; siting avoids overhangs, gutters and splash sources; routine cleaning is part of scheduled maintenance.
Configurable cumulative totals; thresholds ≥ 25 mm/hr, ≥ 45 mm/3 h, ≥ 120 mm/3 days with vigilance timers	Comply (configurable). All accumulation windows, threshold values, alert wording and the 6 h / 12 h / 48 h vigilance-and-reset logic are implemented in the alert engine — see Section 7.

5.2 Flood water level sensing

Proposed products: a **YGRD-65-D terahertz (120 GHz) radar water-level sensor** for continuous, non-contact 24x7 measurement, paired with an **RS PRO RSF80-series external float switch** (both detailed below), at each flood location. Both are mounted on a **dedicated 2.5 m aluminium mast** (with cross-arms), positioned above the water with anti-vandal protection and flexible installation height, and each water-level point carries a clearly marked manual gauge board for verification.

Dual-sensing rationale. The radar sensor provides the accurate continuous level reading; the float switch provides an independent contact confirmation at a set height. The float switch verifies the radar reading and acts as a **backup trigger to start the pump** — for example, when the radar reports water reaching the pump-

start level, the float switch confirms it before (or independently of) the pump starting. If either device fails, the system continues to operate on the surviving output, delivering graceful degradation for this safety-critical measurement. The YGRD-65-D and the RSF80 are specified in detail below.

Manual stick (staff) gauge for verification. In direct response to the Statement of Requirements - which requires "a clearly marked physical gauge for manual verification by maintenance personnel" at each sensor location - Observator will install a **graduated manual stick / staff gauge at every water-level monitoring point**. It is a robust, clearly graduated staff gauge (marine-grade, UV-stable, vandal-resistant) fixed to the 2.5 m mast immediately beside the radar, with its **zero set to the same survey datum as the radar sensor**. Maintenance personnel can then read the water level by eye and compare it directly with the value shown by the radar and on the portal - a simple, power-free field check of sensor accuracy during routine and post-event inspections. Each gauge's datum and reading procedure are recorded in the O&M manual. A gauge is provided at every water-level point (Locations 1-6); both flood points at Location 6 (up and down tunnels) each carry their own. The radar, float and staff gauge at each point are mounted together on the 2.5 m mast.

Gauge plate construction. The flat gauge plates carry metre markings screen-printed in black alkyd enamel on yellow-coated aluminium. The aluminium is H34 5005, and each gauge plate is **0.6 mm x 1000 mm x 130 mm**, with the markings allowing depths to be read at **10 mm increments**. The gauge plates are a proven, manual method of measuring dam and flood-water depths.



Manual staff / stick gauge in service: black alkyd-enamel metre markings screen-printed on a yellow-coated H34 5005 aluminium plate (0.6 x 1000 x 130 mm), reading depth in 10 mm increments, rail-mounted beside the water for manual verification.

Radar water-level sensor - YGRD-65-D

The radar level sensor specified is the **YGRD-65-D terahertz (120 GHz FMCW) radar water-level gauge**, a non-contact instrument mounted above the water on the headwall, pier or a dedicated bracket. It has no moving parts, does not touch the water, and is unaffected by water quality, sediment or floating debris - well suited to floodwater carrying ballast fines, vegetation and litter.



YGRD-65-D terahertz radar water-level sensor (manufacturer render).

Attribute	YGRD-65-D specification and application
Measurement principle	120 GHz terahertz FMCW radar, non-contact, with a narrow 3-degree beam and intelligent false-echo learning - suited to confined channels and sumps.

Attribute	YGRD-65-D specification and application
Range	0.1-50 m (30 m model: 0.1-30 m), far exceeding the 3,000 mm required; blind zone approx. 100 mm.
Accuracy	Plus/minus 1 mm - well within the Statement of Requirements' plus/minus 5 mm or 0.5% tolerance.
Resolution	0.1 mm.
Output / protocol	RS-485 Modbus RTU (4-20 mA and SDI-12 options), interfacing directly with the OMC-048.
Power	12-24 VDC, approx. 40 mA - low draw, suited to the solar / battery station.
Ingress protection	IP67 (IP68 optional); cast-aluminium housing with lens antenna.
Operating temperature	-30 to +80 C.
Pedigree	Tested by China's Ministry of Water Resources hydrological instrument centre; proven on open channels and natural rivers with rapid level change.

Why it is selected. Its non-contact terahertz radar has no moving parts and is immune to sediment, weed and debris fouling - the dominant failure mode for level devices in a gritty rail flood plane - while the narrow 3-degree beam and false-echo learning give clean readings in confined channels and sumps. It interfaces directly with the OMC-048 over RS-485 Modbus RTU and draws only about 40 mA, suiting the autonomous solar station.

Why it is accurate. The FMCW terahertz technique resolves level to 0.1 mm with plus/minus 1 mm accuracy - roughly five times better than the plus/minus 5 mm the Statement of Requirements allows - and the proprietary wave-elimination algorithm holds that accuracy over a disturbed water surface. Its readings are continuously cross-checked against the RSF80 float switch at the float's trip height (Section 5.6).

Why it is efficient. With no moving parts it is effectively maintenance-free and needs no routine cleaning; its low current draw preserves the station power budget; and its RS-485 / Modbus output integrates with the OMC-048 without extra signal conditioning. Combined with the float switch it delivers the graceful-degradation redundancy required for this safety-critical measurement.

Float switch selection - RS PRO RSF80 series

The contact float switch specified is the **RS PRO RSF80-series external vertical float switch**, mounted through the wall of the pump sump or a stilling well via its 1/2 in. NPT thread so it can be installed and serviced without entering the chamber.



RS PRO RSF80-series external vertical float switch (manufacturer photo).

Attribute	RSF80 specification and application
Type and mounting	External vertical float switch; 1/2 in. NPT through-wall mount - no internal chamber access needed to install or service.
Body material	Nylon, Polypropylene, PPS or PVDF options; PPS (or PVDF) selected here for resistance to gritty, sediment-laden floodwater and outdoor exposure.
Contact form	Reversible N/O or N/C by rotating 180 degrees, so the contact is wired for fail-safe behaviour into the OMC-048 input.

Attribute	RSF80 specification and application
Switching rating	240 V AC / 120 V DC max, 0.6 A, 25 VA (resistive) - comfortably drives the OMC-048 volt-free digital input.
Actuation band (SG=1)	Must-close approx. 7-9 mm, must-open approx. 20-24 mm: a tight, repeatable mechanical trip band with built-in hysteresis.
Min. fluid specific gravity	0.85. Floodwater (SG approx. 1.0, higher with sediment) is well above this, guaranteeing reliable flotation.
Temperature range	-20 to +75 C (Nylon) up to -10 to +120 C (PPS) - ample margin for trackside conditions.
Connection	Sealed M12 connector (recommended) or 100 cm flying lead.
Recommended variant	RS PRO RSF80, PPS body, M12 connector (RS stock no. 1919511); Nylon M12 (1748426) as an economical alternative.

Why it is selected. The float switch is deliberately a different technology from the radar: a simple mechanical float with a sealed switch fails in a different way from an electronic radar, so the two together give genuine redundancy with no common-mode failure. Its external through-wall mounting lets maintainers install and replace it from outside the sump - avoiding confined-space entry - and the reversible N/O / N/C contact lets us wire it so that a broken wire or lost contact reads as 'water present' (fail-safe towards pumping). The chemically resistant body suits the ballast fines, vegetation, litter and sediment expected in the corridor, and the low-voltage rating interfaces directly with the OMC-048. It is a professionally approved, widely stocked, low-cost part, which makes spares holding and fast replacement straightforward.

Why it is accurate. As a point-level device its accuracy is switching repeatability, not a continuous plus/minus reading - and that is exactly its role here. The mechanically fixed must-close (approx. 7 mm) and must-open (approx. 20 mm) levels give a precise, repeatable trip point with defined hysteresis, so it reliably confirms the radar at a known height and provides a dependable pump-start point. It does not drift and needs no calibration, and because floodwater specific gravity (approx. 1.0) far exceeds the 0.85 minimum, flotation and actuation stay consistent. The radar continues to provide the continuous, millimetre-resolution measurement; the float provides the trustworthy fixed reference against which the radar is cross-checked - a disagreement at the trip level beyond tolerance raises a discrepancy alarm (Section 5.6).

Why it is efficient. The switch is passive and draws **zero standby power**, which suits the solar / battery autonomous station. It needs no calibration and minimal maintenance, actuates directly and immediately (negligible latency to the pump trigger), and is inexpensive and easy to stock - all of which reduce lifecycle cost and support the availability and response-time obligations.

Requirement	Solution response
Type — non-contact ultrasonic/radar preferred; pressure transducer acceptable	Comply. Non-contact radar (primary) plus contact float switch (verification and backup pump trigger).
Measurement range 0–3,000 mm above datum (min)	Comply. The YGRD-65-D range (0.1-50 m) far exceeds 3,000 mm; mounting height set per location.
Accuracy ± 5 mm or $\pm 0.5\%$ of reading (greater)	Comply. The YGRD-65-D is accurate to ± 1 mm, well within the requirement.
Resolution 1 mm	Comply. YGRD-65-D resolution 0.1 mm.
Continuous monitoring; event-triggered reporting at >50% of warning threshold	Comply (configurable). Radar sampled continuously; event-triggered fast reporting and real-time pump logic run in the OMC-048 logger.
IP68 sensor head; IP66 electronics enclosure	Comply / clarify. The radar head sits above water with an IP66/IP67 enclosure; the YGRD-65-D radar sits above water and is rated IP67 (IP68 optional); the RSF80 float switch is a sealed through-wall device with a sealed M12 connection (immersion rating of wetted parts confirmed for the selected variant); the station electronics enclosure meets IP66. Confirmed per location.

Requirement	Solution response
Manual gauge for verification; anti-vandal fixed mounting	Comply. Marked staff/gauge board at each site; anti-vandal mounting to fixed structures.
Rate-of-rise detection independent of absolute threshold	Comply (configurable). Rate-of-rise alarming configured in addition to absolute thresholds.
Placed at lowest point of flood plane; thresholds & alert wording (standing water / above rail foot / falling)	Comply (configurable). Radar sited to read the lowest point; standing-water, above-rail-foot and trending-down alerts with the specified wording per Annexure C.

5.3 Temperature measurement

Proposed product: a **Gill GMX300 compact temperature and relative-humidity sensor** (which also measures barometric pressure), mounted 1.5-2 m above rail level in a shaded, well-ventilated position away from heat-absorbing surfaces. The GMX300 is a single integrated unit with a built-in radiation screen and no moving parts, and it outputs digitally (SDI-12) to the OMC-048. Because it is mounted independently of the 10 m wind mast it meets the 1.5-2 m temperature-siting requirement directly, and its integrated screen provides sun-shielded temperature and humidity measurement without a separate radiation shield.



Gill MaxiMet GMX300 compact temperature/humidity (and barometric pressure) sensor with an integrated radiation screen (manufacturer image).

Requirement	Solution response
Measurement range –20 °C to +70 °C (min)	Comply (confirm). The GMX300 temperature range covers the operational band; the +70 °C upper limit confirmed against the datasheet.
Accuracy ±0.3 °C or better	Comply (confirm). The GMX300 temperature accuracy is confirmed by calibration certificate; its integrated radiation screen ensures accurate, sun-shielded readings.
Resolution 0.1 °C or better	Comply.
Reporting — configurable, default 1-minute	Comply (configurable). 1-minute logging in the OMC-048 logger.
Ingress protection IP66 (min)	Comply (confirm).
Mounting — radiation shield / louvred enclosure, 1.5 m above rail	Comply. The GMX300 is mounted at 1.5-2 m with its integrated radiation screen, in a shaded position away from concrete, asphalt and building faces, independent of the wind mast.
Thresholds ≥38 °C / ≥45 °C (rising >5 min) and falling logic (>10 min) with TSR and heat-patrol wording	Comply (configurable). Rising/falling thresholds, dwell timers and the 60/40 kph TSR and CJC-T heat-patrol alert wording implemented in the alert engine.

5.4 Wind measurement

Proposed product: the **Gill WindSonic 75 ultrasonic anemometer** (wind speed and direction, no moving parts), mounted at a minimum 10 m above rail level or as specified per location. The WindSonic 75 measures across the full 0-75 m/s (270 km/h) range required by the Statement of Requirements.



Gill WindSonic 75 ultrasonic anemometer (manufacturer photo).

Requirement	Solution response
Type — ultrasonic (preferred) or cup/vane	Comply. Gill WindSonic 75 ultrasonic anemometer (no moving parts).
Wind-speed range 0–75 m/s (min)	Comply. The WindSonic 75 measures 0-75 m/s (270 km/h), meeting the required range in full.
Wind-speed accuracy ± 0.3 m/s or $\pm 3\%$ (greater)	Comply (confirm). Ultrasonic accuracy confirmed per selected model.
Direction range 0–360° continuous; accuracy $\pm 3^\circ$	Comply (confirm).
Averaging — 2-min mean and 3-sec gust (both reported)	Comply (configurable). Both statistics computed and reported.
Ingress protection IP66 (min)	Comply (confirm).
Mounting height ≥ 10 m above rail (or per location)	Comply. On dedicated masts; bridge sites account for wind channelling/local acceleration in siting analysis.
Thresholds ≥ 75 / ≥ 85 kph (SW) and ≥ 120 kph (all external track) with TSR / block-line wording per listed kilometrages	Comply (configurable). Gust-or-mean thresholds and the per-location TSR and block-line alert wording implemented in the alert engine.

5.5 Trackside flood pumps - integration with existing pumps (Marrickville)

The trackside flood pumps at Marrickville **already exist on site**. Observator supplies monitoring and control instrumentation, not pumps, and will **integrate to the existing pumps and their control gear** rather than supply new pumps. The OMC-048 data logger processes the radar level sensor in real time and issues an automatic **start/stop demand to the existing pump control panel**, so the existing pumps power on and off automatically on flood-water detection, with the **contact float switch as an independent backup trigger** should the radar reading be unavailable (see Section 5.6 for the detailed configuration). The OMC-048 also reads the pumps' run and fault status back from the existing panel for alerting and reporting, keeping sensing, control and alerting under single-vendor accountability while re-using the pumps already installed at the site.

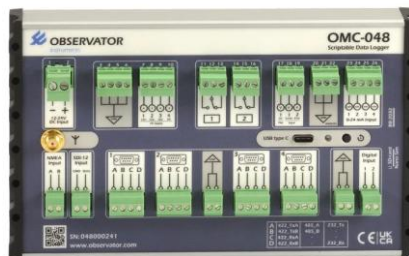
Interfacing to the existing pumps. Integration is by a small, well-defined set of signals between the OMC-048 and the existing pump control panel: volt-free relay outputs from the OMC-048 issue the start/stop demand (the OMC-048 switches the control circuit, not the motor supply), and digital inputs read pump run confirmation and common-fault / motor-protection status. Where the existing pumps expose condition-monitoring outputs (for example a fitted Grundfos IO 113, or winding-temperature and seal alarms), these are also read into the OMC-048 and surfaced on the portal and in alerts. The exact interface terminals, ratings and available status signals are confirmed on site during detailed design (Section 15).

Requirement	Solution response
Pumps at the Marrickville flood site	Existing on site. Observator integrates to the existing pumps and control gear; no pumps are supplied.

Requirement	Solution response
Auto on/off from flood sensors	Comply. The OMC-048 logger starts and stops the existing pumps in real time from the radar level sensor, with the float switch as an independent backup trigger, and alarms on run, fail and no-flow.
Duty / standby operation	Comply (via the existing pumps). The OMC-048 implements lead/lag start levels, alternation and standby cut-in on a duty-pump fault, issued to the existing panel (Section 5.6); the final behaviour is matched to the existing pumps and confirmed with MTS.
Pump status and leading-indicator alarms	Comply. Pump run, fault and (where available) condition signals from the existing panel are read by the OMC-048 and reported on the portal and by screen / email / SMS alerts (Section 3.3.3).
Positioning clear of KE+200 at all times	Comply. Observer equipment (logger cabinet and cabling) is positioned clear of the kinematic envelope +200 mm; the existing pumps are unchanged.
Existing outfall and pipework	Unchanged. The existing pumps, pipework and outfall remain as installed.
Electrical supply	Existing. The existing pumps use their existing supply; the OMC-048 interface is low-voltage and does not switch the motor supply directly. Any panel interface modification is confirmed with MTS.

5.6 Local pump control configuration (OMC-048)

The OMC-048 is a fully **scriptable** data logger. Beyond routine acquisition and telemetry, it executes user-defined control logic **locally and in real time**, independent of the central server and the cellular link. At Marrickville this logic continuously reads the radar water-level sensor and the contact float switch and drives the trackside pump control panel, so the duty and standby pumps operate correctly even if communications are lost. All setpoints, timers and logic below are configuration items held on the OMC-048 and finalised with MTS; none requires a hardware change to adjust (subject to configuration management, Section 14).



Observer OMC-048 scriptable data logger (manufacturer photo).

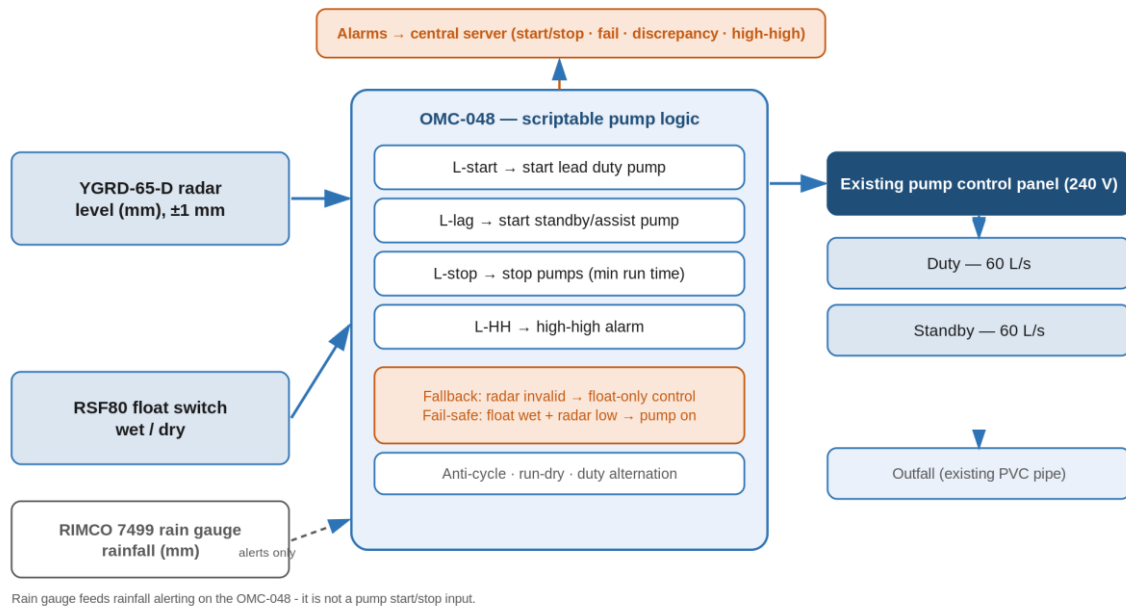


Figure 5 — Marrickville pump control on the OMC-048: radar and float-switch inputs, level setpoints, fail-safe fallback logic, and duty/standby pump outputs. The RIMCO 7499 rain gauge is shown feeding rainfall alerting on the same OMC-048, but is not a pump start/stop input.

Inputs, outputs and interfaces

Signal	Configuration on the OMC-048
Radar water-level sensor (YGRD-65-D)	Wired to the OMC-048 over RS-485 Modbus RTU (SDI-12 / 4-20 mA options also available). Polled every 1-5 s, giving level in millimetres above datum (to ± 1 mm), then range- and plausibility-checked.
Contact float switch (RS PRO RSF80)	Wired to a volt-free digital input, software-debounced, giving a wet/dry state at its fixed trip height; the reversible N/O-N/C contact is set for fail-safe operation.
Duty (lead) pump - start/stop	A volt-free relay/digital output to the pump control-panel contactor (the OMC-048 switches the control circuit, not the 240 V motor supply directly).
Standby / assist pump - start/stop	A second volt-free relay output, energised on rise to the high-high level or on a duty-pump fault; lead/lag alternates each cycle.
Pump feedback from the panel	Digital inputs for running confirmation and motor-protection trip per pump (from the existing pump control panel), plus any available condition alarms/warnings and optional no-flow signals.
Power and enclosure status	Inputs for power supply health and cabinet-door and battery/charge state, used as leading-indicator alarms.

Level setpoints (all configurable, agreed with MTS)

Setpoint	Configured OMC-048 action
L-start (lead)	Rising level reaches the lead-pump start level (sustained beyond a debounce time), then start the lead duty pump.
L-lag (standby / assist)	Level continues to rise to the high-high level, then start the second (standby/assist) pump so both run together.
L-stop	Falling level reaches the stop level, then stop the pumps once the minimum run time has elapsed. A dead-band between L-start and L-stop (hysteresis) prevents rapid cycling.
L-HH (high-high)	High-high level raises a high-high alarm and correlates with the flood 'above rail foot' / block-line alert wording (Section 7).
Float trip height	Independent contact confirmation and backup start/stop point (see below).

Real-time control sequence

- The OMC-048 samples the radar continuously and computes the level in millimetres above datum.
- On a sustained rise to **L-start**, it starts the lead pump and logs and annunciates a 'pump start' event; on a further rise to **L-lag** it starts the second (standby/assist) pump so both run together.
- On a sustained fall to **L-stop**, it stops the pumps, respecting the minimum run time, and logs a 'pump stop' event.
- The **lead/lag assignment alternates each cycle** to equalise wear and run hours across the two pumps; run hours are logged per pump.

Float switch - verification and backup trigger

The float switch gives the OMC-048 a second, independent measure of water presence at a fixed height, and the script uses it three ways:

- **Verification.** In normal operation the float going 'wet' at its trip height cross-confirms the radar reading at the corresponding level; agreement is logged, and a persistent disagreement raises a sensor-discrepancy alarm.
- **Backup trigger.** If the radar becomes invalid - out of range, an SDI-12/Modbus fault, or a frozen value - the OMC-048 automatically reverts to **float-switch-only control**: it starts the pump when the float is wet and stops it (after the minimum run time) when dry, and sends a system-fault alert telling MTS that pump control has moved to the backup input.
- **Fail-safe.** If the float reports 'wet' while the radar is still below L-start (a discrepancy beyond the configured tolerance and time), the logic assumes water is present, starts the pump, and alarms - erring towards pumping rather than risking undetected flooding.

Pump protection, redundancy and alarms

- **Anti-cycling:** the hysteresis dead-band plus minimum-run, minimum-rest and maximum-starts-per-hour timers protect the motors.
- **Run-dry protection:** pumps are inhibited below L-stop.
- **Standby cut-in:** the standby pump starts automatically on loss of a duty pump's running feedback or on a motor-protection trip, sustaining the >99.5% availability target.
- **Immediate alarms:** pump start/stop, pump fail, no-flow, radar/float discrepancy, high-high level and panel/power faults are transmitted immediately - independent of the routine 5-minute telemetry cycle - into the screen/SMS/email alert engine (Section 7).
- **Autonomy:** because the sequence runs on the OMC-048 itself, pumping and local alarming continue through any loss of the central server or cellular link, with buffered data and events forwarded on restoration.

Operators see and supervise all of this through a dedicated station screen in the portal. The Marrickville screen brings together the live sensor readings and station health, the water-level trend against the pump-start and high-high thresholds, the status and condition of the duty and standby pumps (including any condition-monitoring signals available from the existing pumps), and a supervised manual start/stop control for the pumps.

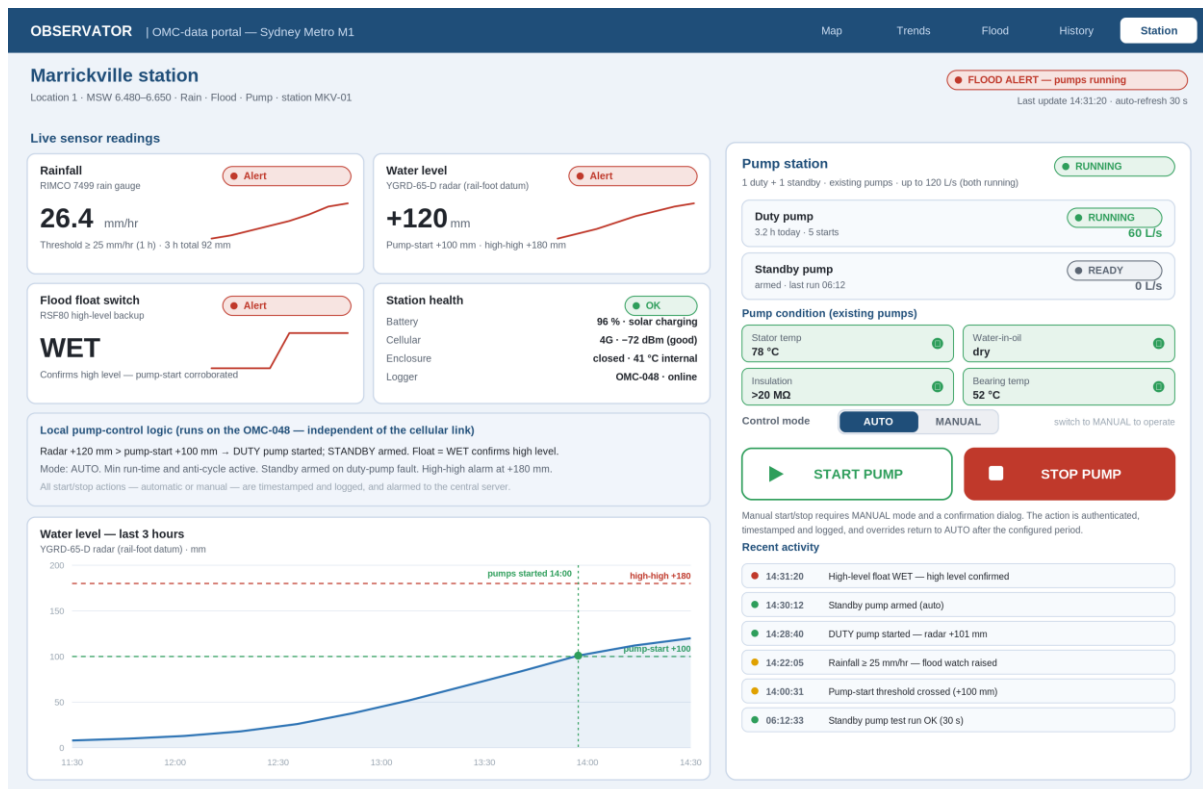


Figure 12 - Marrickville pump-station screen: all live sensor readings and station health, the water-level trend against the pump-start (+100 mm) and high-high (+180 mm) thresholds, duty/standby pump status with existing-pump condition monitoring, an AUTO/MANUAL mode toggle and a supervised START/STOP pump control - manual operation requires MANUAL mode plus a confirmation dialog, and every action is authenticated, timestamped and logged (demonstration data).

5.7 Station infrastructure - mast, enclosure and solar power system

Each station is supplied as a complete, self-contained package. Beyond the sensors, the balance-of-plant - the monitoring mast, the equipment enclosure and the solar power system - is engineered by Observator for the trackside environment: corrosion-resistant, low-maintenance, safe to service, and sized for full autonomy with no mains power or fixed communications on site.

Monitoring mast (VM5F telescopic)

The anemometer is carried on a dedicated multi-section telescopic mast - the VM5F (VM5-11.3) - raised to the standard wind-exposure height called for in the Statement of Requirements (the mast extends to 11.3 m, so wind can be set at approximately 10 m), while the temperature/humidity sensor and rain gauge are mounted lower on the same structure at 1.5-2 m.



VM5F (VM5-11.3) multi-section telescopic mast with self-locking hand winch and base (manufacturer image).

- **Construction.** VM5F VM5-11.3 multi-section retractable steel mast - five telescoping square-tube sections (2.5 m each, tapering from a 60 mm top section to a 140 mm base) giving an extended height of 11.3 m and a retracted height of approximately 2.74 m. Anti-corrosion protection is hot-dip galvanising plus powder coat, and the mast is wind-rated to 120 kph; wind loading is confirmed to AS/NZS 1170.2 for the Sydney region at detailed design.
- **Safe maintenance.** The mast is raised and lowered on a self-locking hand winch (1800 lb; an electric 4500 lb DC 12/24 V winch or a 2.2 kW hydraulic cylinder are available options), so the anemometer and upper equipment can be lowered to approximately 2.74 m and serviced safely at ground level - avoiding work-at-height, elevating work platforms and extended rail possessions.
- **Guying and stability.** The mast is stabilised by three approximately 14 m tensioning guy ropes (6 mm diameter), or a self-supporting cable option per the VM5F selection table, sized to the site and the 120 kph wind rating.
- **Foundation.** Concrete footing or engineered ballasted base to suit each sites geotechnical conditions and access constraints.
- **Lightning and earthing.** Integral lightning air-terminal and earthing/bonding to AS 1768, with surge protection carried into the enclosure.
- **Cabling.** Internal cable way and sensor cross-arms set each sensor at the height the Statement of Requirements specifies (wind at 10 m; temperature/RH at 1.5-2 m).
- **2.5 m masts at the water-level points.** At each water-level monitoring point the YGRD-65-D radar, RSF80 float switch and manual staff gauge are mounted together on a **2.5 m aluminium mast fitted with cross-arms** - a lighter, corrosion-resistant structure. At the flood-only sites this mast also carries the OMC-048 enclosure, solar panel, battery and charge controller. **Seven** 2.5 m masts are required - one at each

water-level point (Locations 1-5 and both units at Location 6). The taller telescopic VM5F is used only where a wind sensor requires anemometer exposure (Locations 2, 5 and 7).

Stainless-steel equipment enclosure (OMC-048)

All station electronics are housed in a single, secure, weatherproof enclosure mounted on the mast at a safe working height.

- **Material and rating.** 316-grade stainless-steel enclosure, IP66 (dust- and water-tight), UV-stable and corrosion-resistant for the trackside, coastal-influenced environment.
- **Contents.** Houses the OMC-048 data logger, solar charge controller, battery, cellular modem, terminal rails and surge protection in one integrated, serviceable assembly.
- **Security.** Lockable, tamper- and vandal-resistant, keeping equipment inaccessible to the public as required.
- **Thermal and sealing.** External sun-shade and ventilation manage internal temperature; a sealed gland plate provides weatherproof cable entry.
- **Mounting.** Pole-mounted at approximately 1.3 m (or on an adjacent plinth) for safe, ground-level access.

Solar charge controller

Charging is managed by a dedicated solar charge controller that both protects the battery and reports power-system health back to the logger.

- **MPPT efficiency.** A maximum-power-point-tracking (MPPT) controller, sized to the PV array and 12 VDC battery, maximises energy harvest in winter and low-light conditions.
- **Battery protection.** Temperature-compensated multi-stage charging and low-voltage load disconnect protect and prolong battery life.
- **Monitored.** DIN-rail mounted in the enclosure; PV input, charge state and battery voltage are read by the OMC-048 and alarmed as leading indicators of power-system health (Section 3.3.3).

Battery and autonomy

Each station carries a LiFePO₄ (lithium iron phosphate) battery sized to ride through extended periods of poor solar and any charging fault.

- **Type.** LiFePO₄ (lithium iron phosphate), maintenance-free deep-cycle battery, 12 VDC, requiring no venting or electrolyte maintenance.
- **Autonomy.** Sized for at least 3-4 days (confirmed at design) of operation with no solar input, so monitoring and alerting continue through prolonged overcast weather or a panel fault.
- **Sizing.** The PV array is sized against Sydneys winter solar resource and the station load so the battery fully recharges between events.
- **Monitored.** Battery voltage and state of charge are continuously logged, with low-battery alarms raised into the alert engine.

Together these items deliver a rugged, autonomous, low-maintenance station requiring no mains power or fixed communications on site. The physical arrangement is shown in the station elevations (Section 6.2) and the autonomous-power summary in Section 8.6.

6. Station configuration by location

The sensor mix below follows the matrix in Section 3.3.1 of the Statement of Requirements. Kilometrages and GPS positions will be confirmed with MTS during site visits.

Loc.	Site	Rain	Flood	Temp	Wind	Pump
1	Marrickville Station	✓	✓			✓
2	Between Marrickville & Dulwich Hill		✓	✓	✓	
3	Canterbury station		✓			
4	Campsie station		✓			
5	Belmore triangle	✓	✓	✓	✓	
6	Lady Game Drive (tunnel)		✓ x2 (up + down tunnels)			
7	Windsor Road Single Span Crossing			✓	✓	

Location 1 (Marrickville) hosts the existing trackside pumps, to which the OMC-048 integrates for automatic control and monitoring. Location 6 (Lady Game Drive tunnel) is fitted with two independent flood-monitoring points - one in the up tunnel and one in the down tunnel - each comprising a YGRD-65-D radar level sensor and an RSF80 float switch.

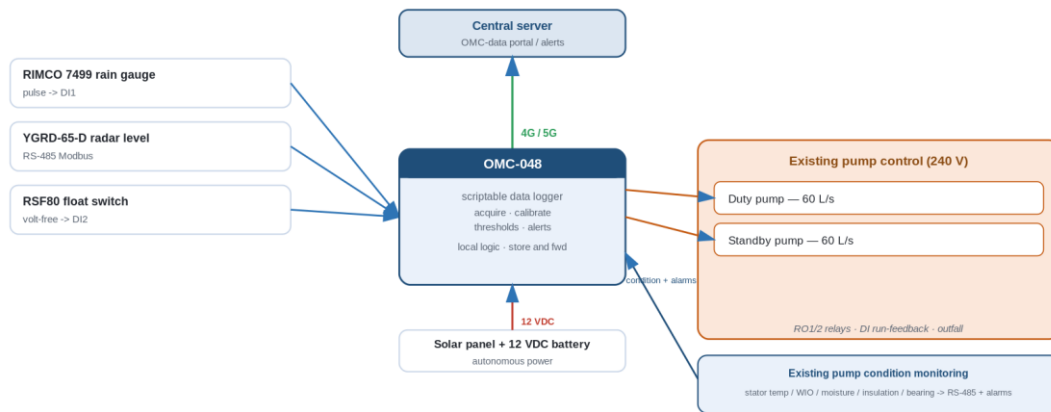
Two data loggers at Location 6. The up-tunnel and down-tunnel sensors at this location can be separated by a considerable distance along the alignment, and running a single long sensor cable between the two points and back to one logger would be costly, difficult to route and install through the tunnel environment, and would introduce a single point of failure. Observator therefore proposes that each of the two water-level sensors be installed as a fully self-contained monitoring unit - its own OMC-048 data logger, its own solar panel and its own battery - sited close to the sensor it serves. This keeps every cable run short and local, removes long in-tunnel cabling, and improves resilience: each unit acquires, validates and transmits its own data over its own cellular link and runs on its own power, so a fault affecting one unit does not disable the other, and both tunnels continue to be monitored and alarmed independently. Both loggers report to the same central server and are presented together on the portal as Location 6.

6.1 Individual station architectures

The diagrams below show the architecture of each station - the sensors fitted, their interface to the OMC-048 data logger, autonomous solar power, the 4G/5G link to the central server, and (at Marrickville) the trackside pump control.

Station 1 — Marrickville

MSW 6.480–6.650 · monitors: Rain · Flood · Pump

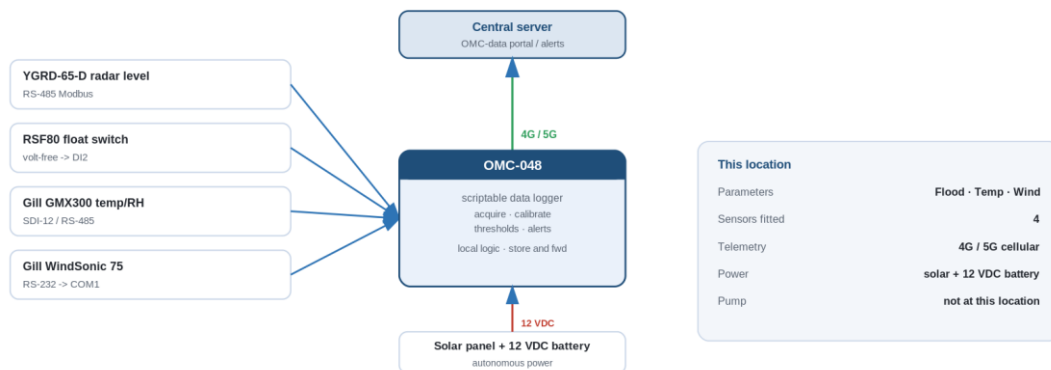


Per-station architecture (schematic). Sensor mix per Statement of Requirements Section 3.3.1 / Annexure C.

Station 1 - Marrickville (Loc 1): rain (RIMCO 7499), radar level + RSF80 float, and the existing trackside pumps (integrated for automatic control).

Station 2 — Between Marrickville & Dulwich Hill

MSW 7.000–8.430 · monitors: Flood · Temp · Wind

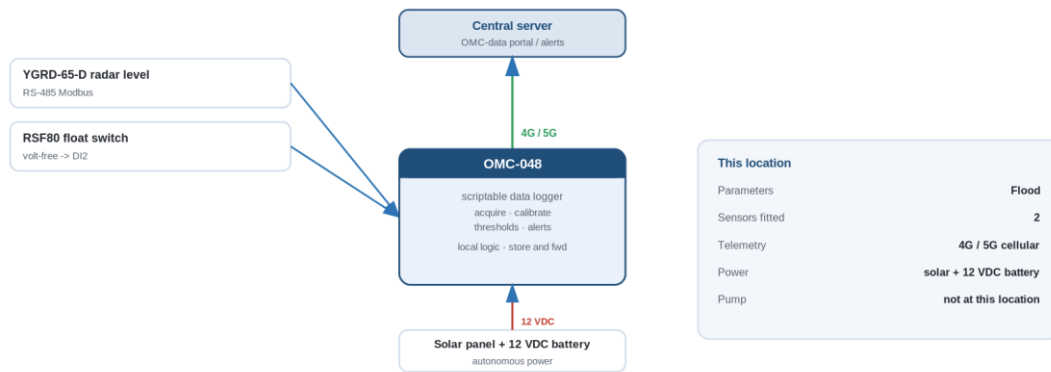


Per-station architecture (schematic). Sensor mix per Statement of Requirements Section 3.3.1 / Annexure C.

Station 2 - Between Marrickville and Dulwich Hill (Loc 2): radar level + float, Gill GMX300 temperature/humidity and WindSonic 75 (wind).

Station 3 — Canterbury station

MSW 10.070–10.550 · monitors: Flood

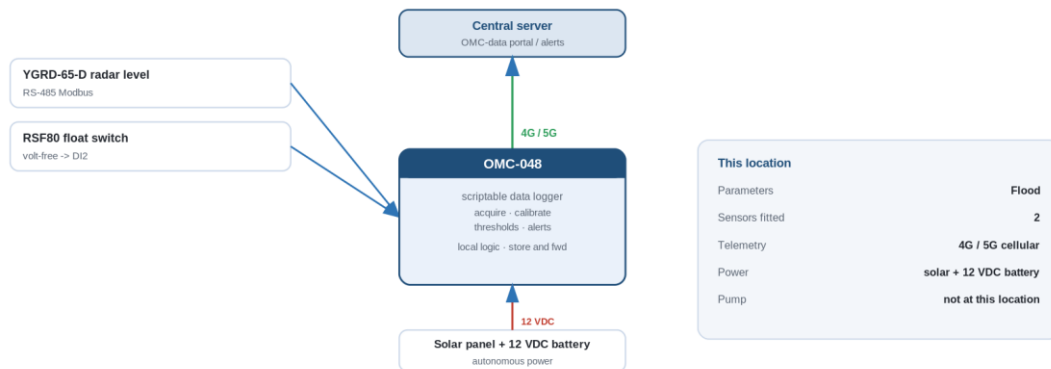


Per-station architecture (schematic). Sensor mix per Statement of Requirements Section 3.3.1 / Annexure C.

Station 3 - Canterbury (Loc 3): radar level + float switch (flood monitoring only).

Station 4 — Campsie station

MSW 11.800–12.460 · monitors: Flood

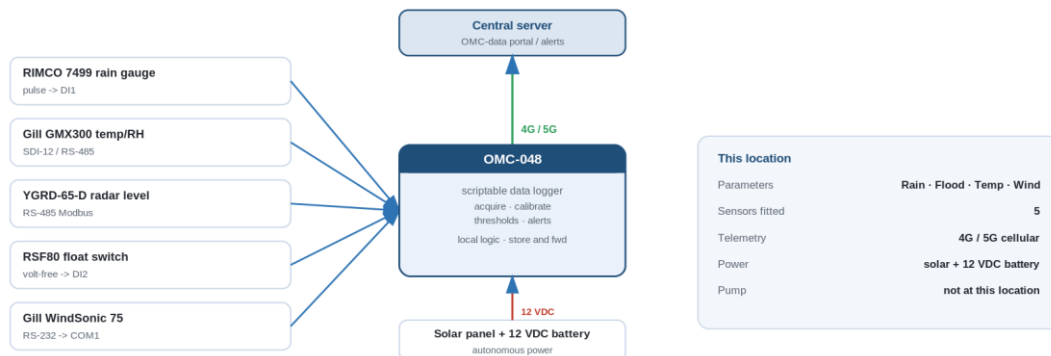


Per-station architecture (schematic). Sensor mix per Statement of Requirements Section 3.3.1 / Annexure C.

Station 4 - Campsie (Loc 4): radar level + float switch (flood monitoring only).

Station 5 — Belmore triangle

MSW 12.400–13.320 · monitors: Rain · Flood · Temp · Wind



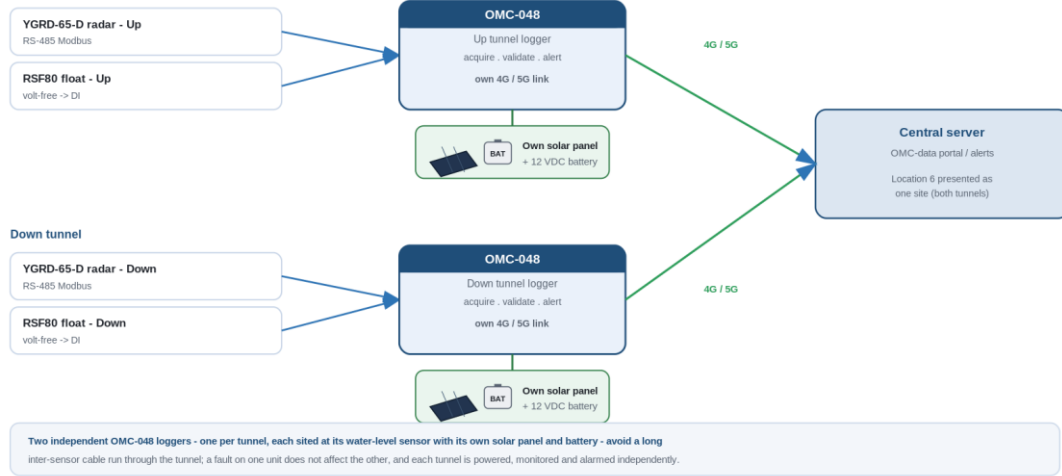
Per-station architecture (schematic). Sensor mix per Statement of Requirements Section 3.3.1 / Annexure C.

Station 5 - Belmore triangle (Loc 5): full suite - RIMCO 7499 rain, Gill GMX300 temperature/humidity, radar + float, WindSonic 75.

Station 6 - Lady Game Drive (tunnel)

tunnel — km confirmed on site . monitors: Flood x2 (up + down tunnels)

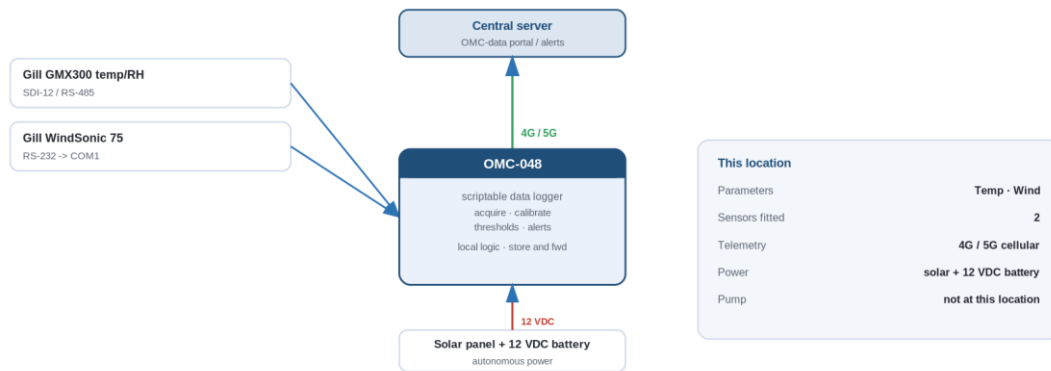
Up tunnel



Station 6 - Lady Game Drive tunnel (Loc 6): two independent flood-monitoring points (up and down tunnels), each with a YGRD-65-D radar and RSF80 float served by its own dedicated OMC-048 data logger and its own solar panel and battery - a fully self-contained monitoring unit.

Station 7 — Windsor Road Single Span Crossing

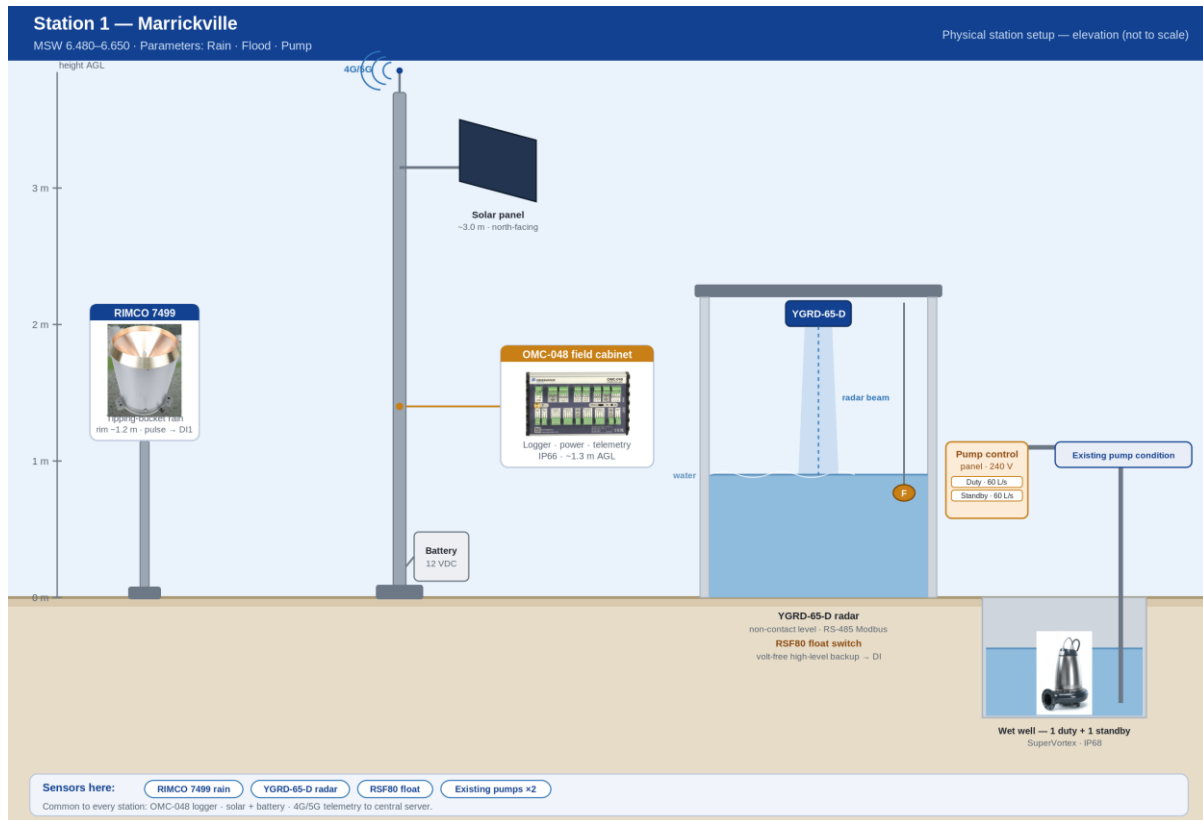
MNW 16.200–16.600 . monitors: Temp . Wind



Station 7 - Windsor Road Single Span Crossing (Loc 7): Gill GMX300 temperature/humidity and WindSonic 75 (wind) - no water-level sensor is required at this location.

6.2 Physical station setup (site elevations)

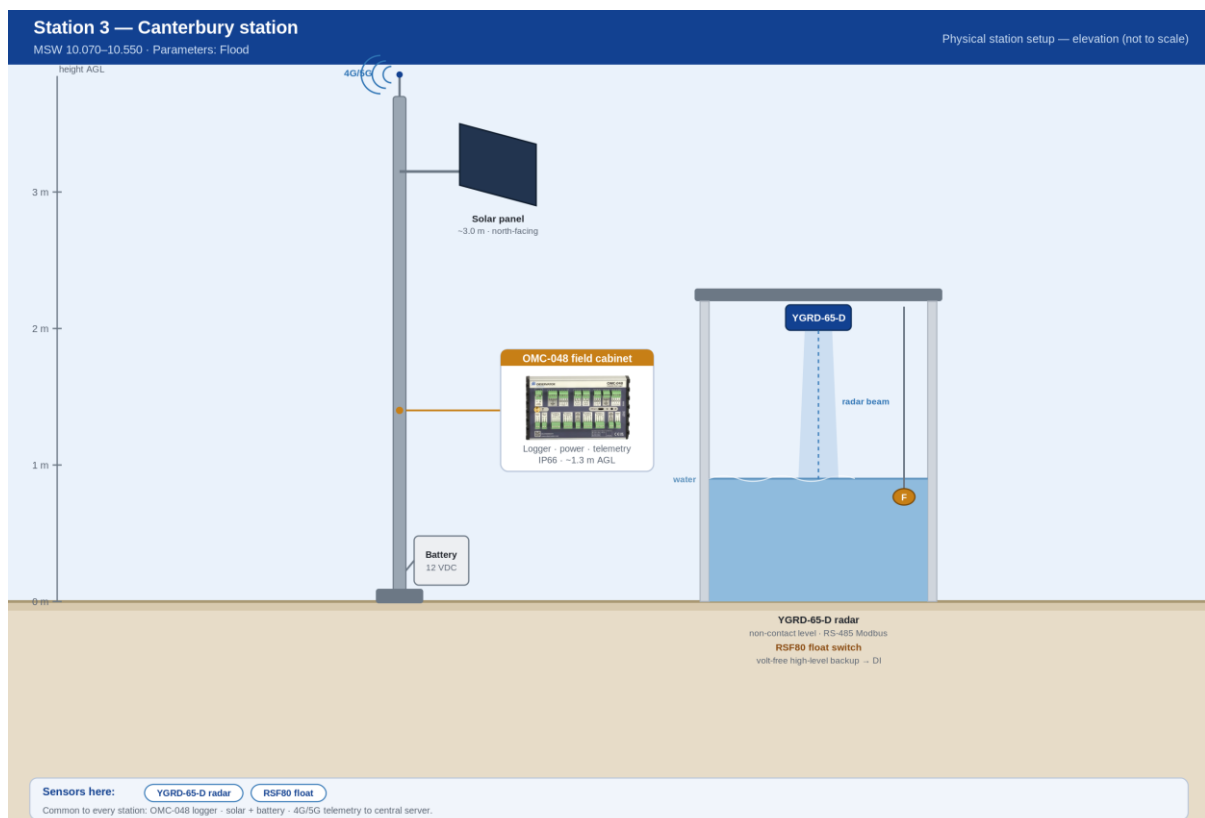
The elevation views below show how each station is physically assembled on site: the mast-mounted meteorological sensors at their approximate heights above ground level (AGL), the north-facing solar panel and battery, the IP66 field cabinet housing the OMC-048, and the flood installation with the non-contact YGRD-65-D radar mounted above the channel and the RSF80 float switch providing high-level backup. At Marrickville the existing submersible pump wet-well and control panel are shown. These are indicative elevations, not to scale.



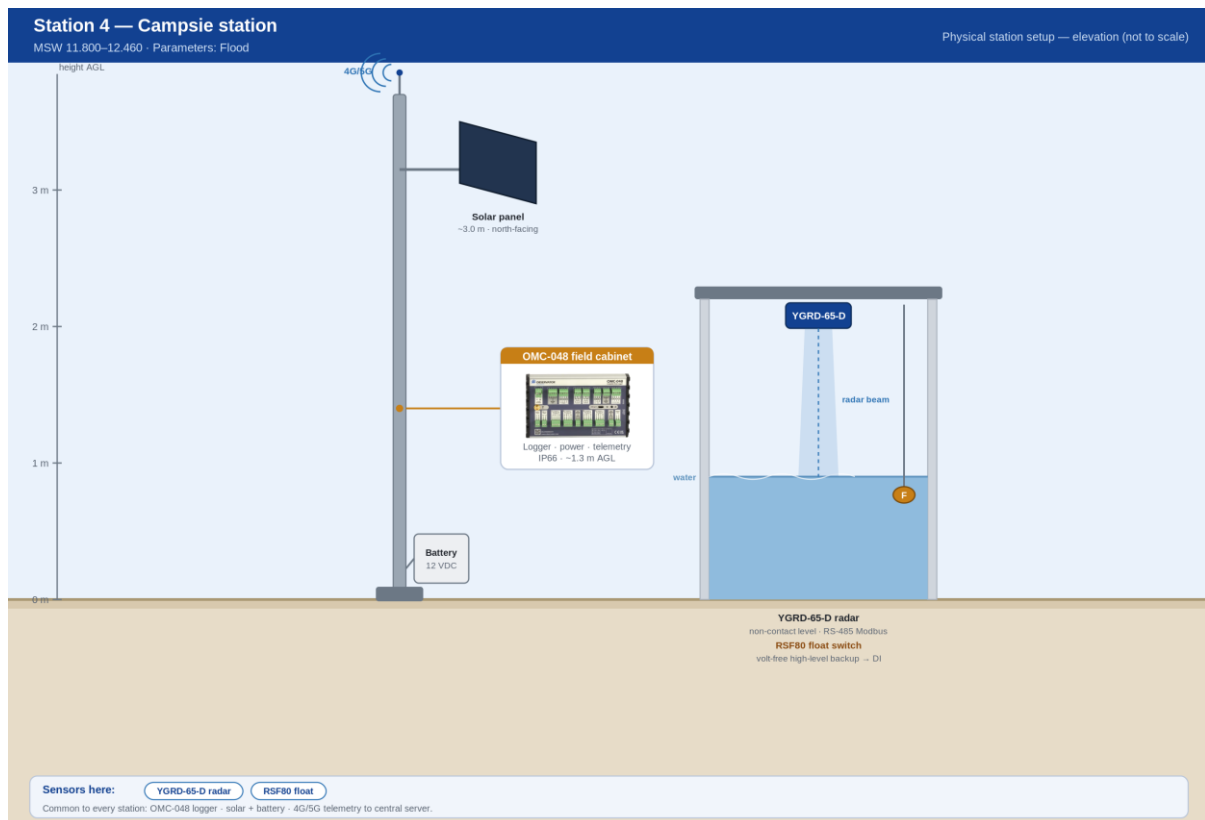
Station 1 - Marrickville: mast-mounted RIMCO 7499 rain gauge, OMC-048 cabinet, radar + RSF80 float over the channel, and the existing submersible pump wet-well with its control panel (integrated to the OMC-048).



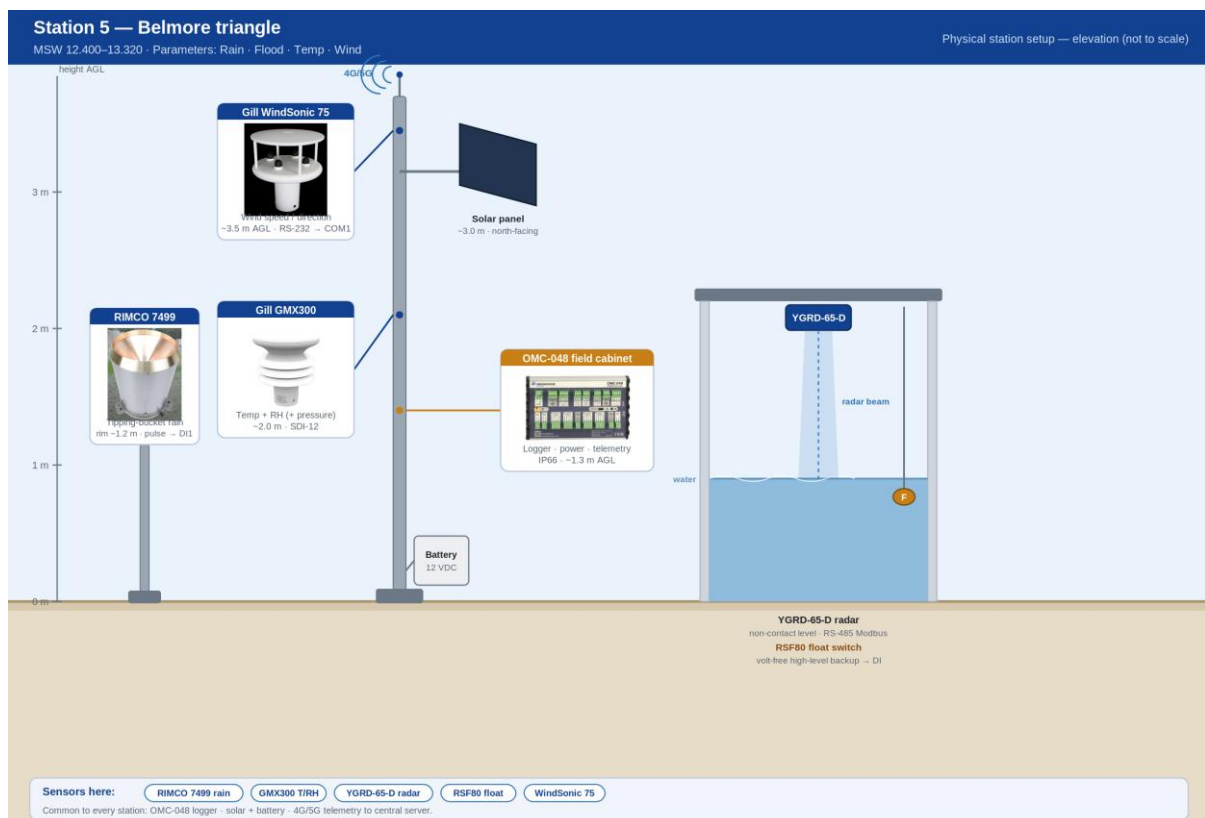
Station 2 - Between Marrickville and Dulwich Hill: WindSonic 75 and the Gill GMX300 temperature/humidity sensor on the mast, OMC-048 cabinet, and radar + RSF80 float over the channel.



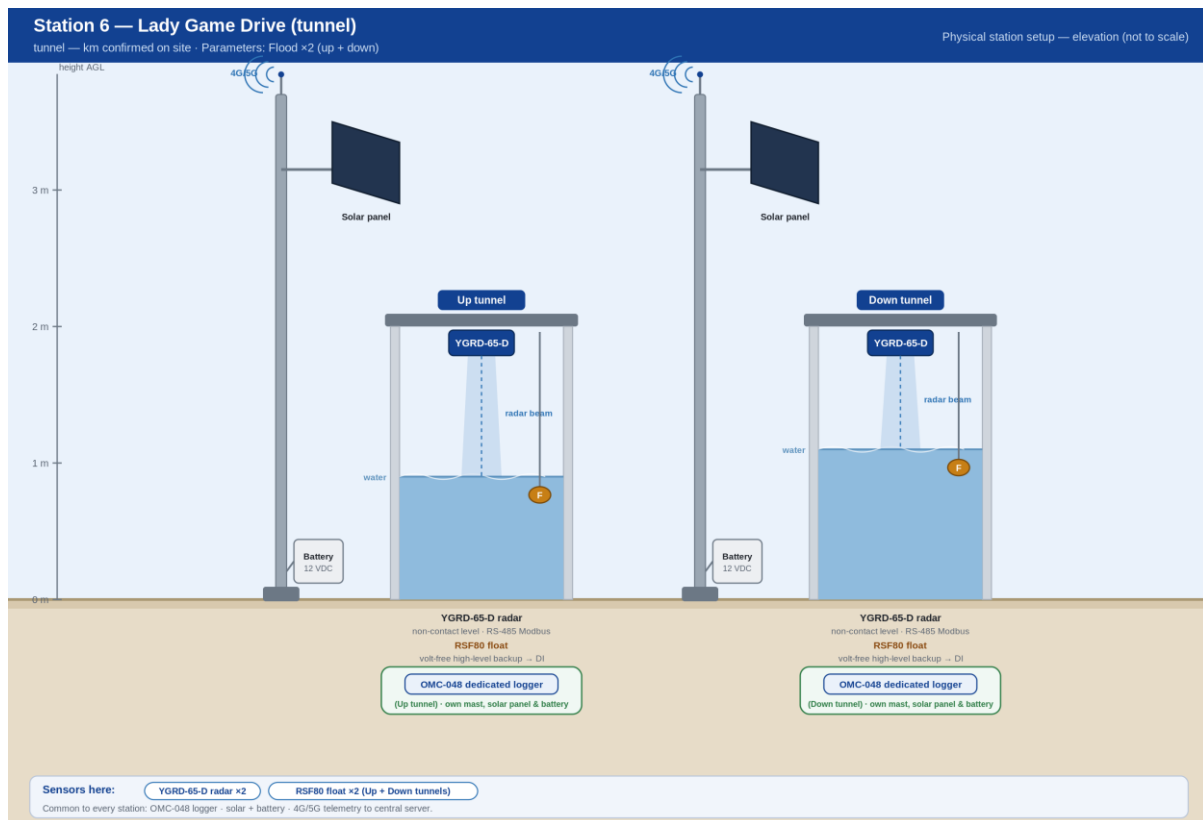
Station 3 - Canterbury: flood monitoring only - YGRD-65-D radar above the channel with RSF80 float high-level backup, OMC-048 cabinet, solar and battery.



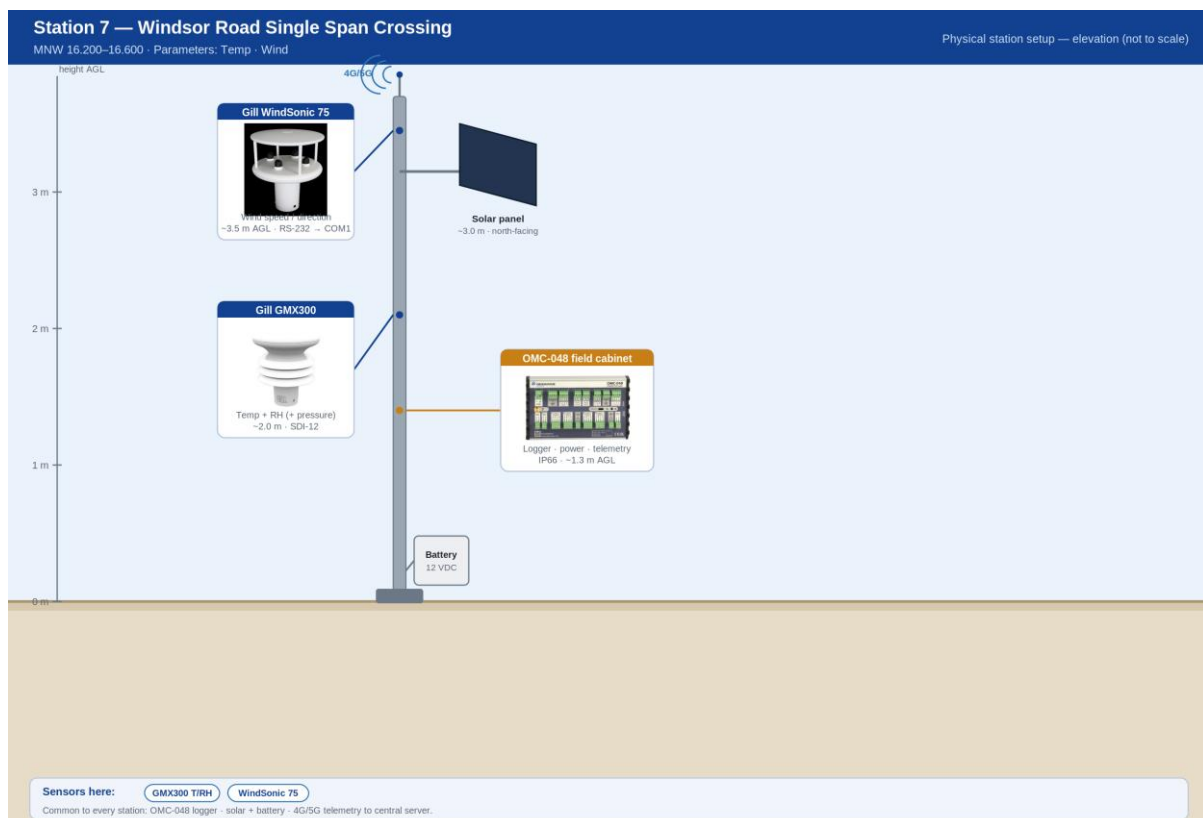
Station 4 - Campsie: flood monitoring only - YGRD-65-D radar above the channel with RSF80 float high-level backup, OMC-048 cabinet, solar and battery.



Station 5 - Belmore triangle: full suite - RIMCO 7499 rain gauge, Gill GMX300 temperature/humidity sensor and WindSonic 75 on the mast, OMC-048 cabinet, and radar + RSF80 float over the channel.



Station 6 - Lady Game Drive tunnel: two channels (up and down tunnels), each with a YGRD-65-D radar and RSF80 float and, sited at the sensor, its own dedicated OMC-048 data logger, solar panel and battery - a fully self-contained monitoring unit that avoids a long inter-sensor cable run.



Station 7 - Windsor Road Single Span Crossing: WindSonic 75 and the Gill GMX300 temperature/humidity sensor on the mast, with the OMC-048 cabinet - no water-level sensor is required at this location.

7. Alerting and threshold logic

The alerting engine is the safety-critical heart of the system. It is implemented partly on the OMC-048 scriptable data logger at each station (for immediate, connectivity-independent detection and real-time pump control) and partly on the central server (for aggregation, map display, distribution and acknowledgement). Every threshold, dwell timer, vigilance period and alert message is a configuration item, finalised with MTS where the Statement of Requirements marks values as "pending confirmation".

Meeting the 5-minute alert obligation. Routine telemetry is transmitted on a configurable cycle (default every 5 minutes). Critically, a threshold breach does not wait for the next cycle: the OMC-048 raises an **immediate, event-triggered transmission** the moment a threshold is crossed, so the screen, email and SMS alerts are issued well within the required 5 minutes.

7.1 Delivery and content

- **Weather alerts** are sent by **screen, email and SMS within 5 minutes** of a threshold being breached.
- **System-fault alerts** are sent by **screen, email and SMS within 30 minutes** of a fault being detected.
- Each alert carries: date identified; time identified; track (up/down) and rail (up/down); the kilometrage of the weather station; and a **hyperlink** to the event on the server.

7.2 Rainfall vigilance logic

For each of the ≥ 25 mm/hr, ≥ 45 mm/3 h and ≥ 120 mm/3-day thresholds the engine issues the specified alert (initiate CJC-T front-of-train patrol between Hurlstone Park and Bankstown; be prepared for network flooding), then:

- keeps the alert **active for the vigilance period** (6 h / 12 h / 48 h respectively) measured from when rainfall falls back below the threshold;
- includes the reminder to demobilise patrols when visibility is limited and to implement a 60 kph TSR between Hurlstone Park and Bankstown until cancelled or visibility improves;
- issues a **cancellation alert** at the end of the vigilance period to prompt restoration of normal operations; and
- **extends and resets** the vigilance period if a further qualifying rainfall event occurs within the window.

How the server implements this — in plain terms

Behind the scenes the server does this with three simple ideas working together: a **running total**, a **countdown timer** and a set of **adjustable settings**. No one has to watch a screen or run a report for it to work — it happens automatically, around the clock.

- **A running total that is always up to date.** As readings arrive from the rain gauge, the server keeps a live tally of how much rain has fallen in the **last hour, the last 3 hours and the last 3 days**. Think of it as a tally board the server updates with every new reading, automatically dropping off rain that has aged out of each window. It constantly compares each tally against its line in the sand (25 mm in an hour, 45 mm in 3 hours, 120 mm in 3 days).
- **Raise the alert the instant a line is crossed.** The moment any tally goes over its threshold, the server issues the alert straight away - with the exact wording MTS requires (initiate the CJC-T front-of-train patrol between Hurlstone Park and Bankstown, be prepared for network flooding) - rather than waiting for the next routine update.
- **Stay cautious after the rain eases (the countdown timer).** When the rain falls back below the line, the server does **not** stand down immediately - because the flooding risk lingers after the rain stops. Instead it starts a countdown (**6 hours, 12 hours or 48 hours** depending on which threshold was hit) and keeps the alert and its instructions in force the whole time the clock is running.

- **Reset the clock if bad weather returns.** If another qualifying burst of rain occurs while the countdown is still running, the server simply **restarts the full countdown from the beginning**, so vigilance never lapses during an ongoing weather event.
- **Sound the all-clear automatically.** When the countdown finally reaches zero with no further qualifying rain, the server sends a **cancellation (all-clear) alert** so normal operations can resume, and records the entire sequence in the event history.
- **Everything is a setting, not fixed in the software.** The threshold amounts, the accumulation windows and the countdown durations are all values the server reads from a configuration screen. MTS can review and fine-tune them at any time without any software change.

For safety, the same logic also runs on the OMC-048 data logger at the station itself, so detection and alerting continue even if the link to the server is briefly interrupted; the logger holds its readings and passes them to the server when the connection returns.

7.3 Flood, temperature and wind logic

- **Flood:** standing-water (PTZ-camera verification), water-above-rail-foot (block the line), and trending-down (system checks, track inspection, staged reinstatement at 25 → 60 kph → unrestricted) alerts, per Annexure C locations.
- **Temperature:** $\geq 38^\circ\text{C}$ and $\geq 45^\circ\text{C}$ rising thresholds (>5 min) and $<45^\circ\text{C}$ / $<38^\circ\text{C}$ falling thresholds (>10 min) driving the 60/40 kph TSR and CJC-T heat-patrol wording.
- **Wind:** ≥ 75 / ≥ 85 kph (Southwest) TSR wording at the listed MSW kilometrages and ≥ 120 kph block-line wording on all external (non-tunnel) track.

Configurable threshold table and rules-based engine

All of the above triggers are held in a single **configurable threshold (alert/warning) table** within the system rather than being written into the software. On award, this table is **pre-populated with the values from the Statement of Requirements and then finalised with MTS** during detailed design - the Statement of Requirements marks a number of values as "pending confirmation", and those cells are simply confirmed or adjusted with MTS before commissioning. Each row defines one alert or warning rule: the parameter and location it applies to, the trigger condition (value and, where relevant, the accumulation window), any dwell/vigilance and reset times, the severity (warning or alert), and the exact wording and recipients of the resulting message.

A **rules-based engine** sits on top of this table. Nominated **MTS administrators can review, change and update the rules** - threshold values, warning-versus-alert levels, dwell and vigilance times, which locations a rule covers, and the message wording and distribution list - through a simple configuration screen, **without any software change or vendor involvement**. New rules can be added and existing ones enabled, disabled or scheduled. Changes are validated, version-controlled and recorded against the user who made them, giving a clear audit trail, and take effect across the network once saved. This keeps the system flexible as MTS refines its operating thresholds over the life of the contract.

The table below is **illustrative of the structure only**; the final parameters, values and wording are those confirmed with MTS.

Parameter / rule	Trigger condition (example — pending MTS confirmation)	Dwell / vigilance	Severity & action
Rainfall — intensity	≥ 25 mm in 1 h (rolling)	Vigilance 6 h; reset on re-trigger	Alert — CJC-T front-of-train patrol; prepare for flooding
Rainfall — short accumulation	≥ 45 mm in 3 h (rolling)	Vigilance 12 h; reset on re-trigger	Alert — patrol; 60 kph TSR until cancelled / visibility improves

Parameter / rule	Trigger condition (example — pending MTS confirmation)	Dwell / vigilance	Severity & action
Rainfall — multi-day	≥ 120 mm in 3 days (rolling)	Vigilance 48 h; reset on re-trigger	Alert — network flooding vigilance
Flood — standing water	Radar level above set point	Immediate	Alert — PTZ-camera verification prompt
Flood — above rail foot	Level above rail-foot set point	Immediate	Alert — block the line
Temperature — heat	≥ 38 °C / ≥ 45 °C rising (> 5 min)	Falling < 38 °C / < 45 °C (> 10 min)	Warning / Alert — 60 / 40 kph TSR; CJC-T heat patrol
Wind — gust	≥ 75 / ≥ 85 kph at listed kilometrages	Per rule	Warning / Alert — TSR wording
Wind — extreme	≥ 120 kph (external, non-tunnel track)	Per rule	Alert — block the line

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Map Trends Alerts Admin - Rules

Alert & warning rules — configuration

The rules-based engine: thresholds, timers, severities and messages are settings — not code. MTS administrators can add, edit, enable or disable any rule.

+ New rule All Rainfall Flood Temperature Wind

Search rules...

Rule / parameter	Trigger condition	Dwell / vigilance	Severity	On	Edit
Rainfall — intensity Rainfall parameter	≥ 25 mm / 1 h (rolling)	Vigilance 6 h - reset	Alert	On	Edit
Rainfall — short accum. Rainfall parameter	≥ 45 mm / 3 h (rolling)	Vigilance 12 h - reset	Alert	On	Edit
Rainfall — multi-day Rainfall parameter	≥ 120 mm / 3 days	Vigilance 48 h - reset	Alert	On	Edit
Flood — standing water Flood parameter	Level > set point	Immediate	Alert	On	Edit
Flood — above rail foot Flood parameter	Level > rail-foot SP	Immediate	Alert	On	Edit
Temperature — heat 1 Temperature parameter	≥ 38 °C rising > 5 min	Fall < 38 °C > 10 min	Warning	On	Edit
Temperature — heat 2 Temperature parameter	≥ 45 °C rising > 5 min	Fall < 45 °C > 10 min	Alert	On	Edit
Wind — gust Wind parameter	≥ 75 / 85 kph (SW)	Per rule	Warning	On	Edit
Wind — extreme Wind parameter	≥ 120 kph (external)	Per rule	Alert	disabled	Edit

Edit rule — Rainfall (intensity)

Rule name: Rainfall — intensity

Parameter: Rainfall Severity: Alert

Trigger condition: 25 mm in 1 hour (rolling)

Dwell: — Vigilance: 6 hours Reset on re-trigger: On

Applies to (locations): Belmore triangle, Marrickville, Canterbury + add

Alert message (wording sent to recipients): Initiate CJC-T front-of-train patrol between Hurlstone Park and Bankstown. Be prepared for network flooding. Demobilise patrols and apply 60 kph TSR if visibility is limited.

Recipients & channels: Ops controllers, CJC-T, Screen, Email, SMS

Pending MTS confirmation Save changes Cancel

All changes are validated, version-controlled and logged. Last change: A. Nguyen (MTS Administrator) - 24 Jul 2026 14:02 - rule set v7. 9 rules - 8 enabled

Figure 17 - Rules-based engine (configuration screen): each alert/warning rule is a row an MTS administrator can edit - parameter, trigger value and accumulation window, dwell, vigilance and reset, severity (warning/alert), the locations it applies to, the exact message wording and the recipients/channels - with per-rule enable/disable toggles and a "New rule" action. All values are pending confirmation with MTS, and every change is validated, version-controlled and logged (demonstration data).

7.4 Redundancy and fallback

On sensor failure the system automatically generates alerts from the designated alternate source and rewrites the alert wording to state which additional section(s) it now covers, per Section 3.3.3 of the Statement of Requirements — for example rain-gauge fallback between Belmore triangle and Marrickville station; flood-sensor fallback to CIDS PTZ monitoring; temperature fallback to Belmore triangle, Bankstown airport (BOM) and Richmond (BOM); wind fallback between Belmore triangle and Marrickville station; and up/down tunnel cross-reading at Location 6. A fault alert first notifies MTS that alerting has switched to the fallback source.

7.5 PTZ camera verification link

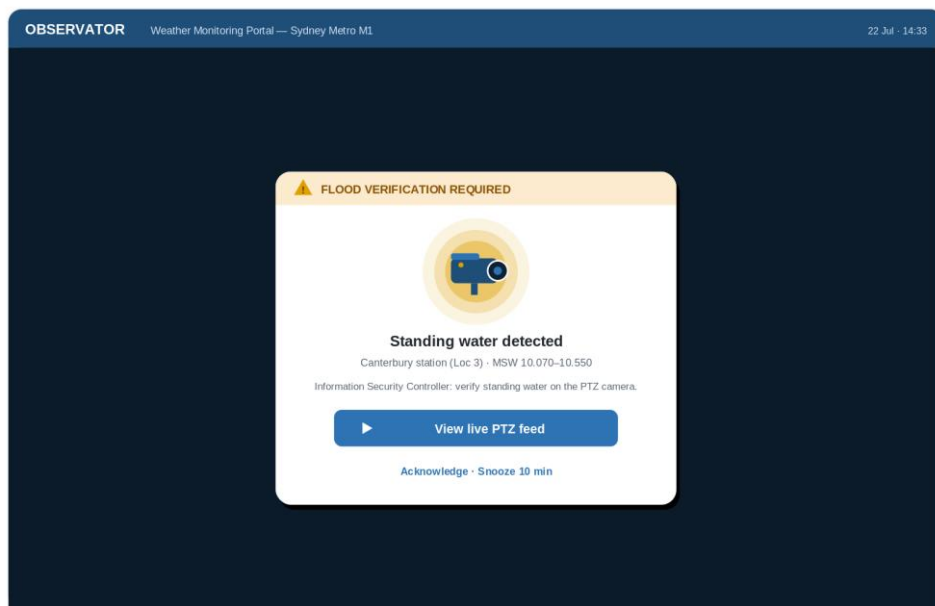
For the “standing water detected” flood condition the Statement of Requirements calls for the Information Security Controller to verify the situation on a Pan-Tilt-Zoom (PTZ) camera before further action. The system

supports this directly: when standing water is detected at a location, the portal **pops up an alert prompting the user to check the PTZ camera**, and the alert carries a **direct, clickable link that opens the live camera feed** for that location - so the operator can view the feed in one click rather than having to locate the camera manually.

- **On screen (portal):** the alert appears as a pop-up on the operator's dashboard with a “View live PTZ feed” button that opens the camera stream for the affected location.
- **On the map / stations view:** the affected station is flagged with a small **red, blinking camera icon** the moment its threshold is reached, drawing the operator's attention; clicking the icon opens the live PTZ feed for that location.
- **By email:** the alert email includes the same hyperlink, opening the live feed in the recipient's browser.
- **By SMS:** the text message includes a short URL to the live feed, openable on a phone or tablet.

The correct camera link is selected automatically from the alarm location (per Annexure C), so the operator is taken straight to the relevant view. This applies to the standing-water alert and can be extended to any location where a camera view aids verification.

Assumption: this capability assumes each relevant PTZ camera exposes a web (URL) link to its live feed that can be embedded in the alerts across screen, email and SMS. The specific camera URLs (and any authentication or network access required to reach them) are provided and confirmed by MTS during detailed design; see Section 15.



PTZ verification pop-up (illustrative). The camera icon blinks to prompt the operator; “View live PTZ feed” opens the live camera stream for the alarm location (link also included in the email and SMS alerts).

Sample screen — PTZ verification pop-up: a blinking camera icon prompts the operator to check the camera, and a one-click “View live PTZ feed” button opens the live stream for the alarm location (the same link is included in the email and SMS alerts).

7.6 Alert and notification history

Beyond the live pop-ups, the portal keeps a complete, searchable history of every event and alert the system has generated. A dedicated alerts and notifications panel lets operators review, filter and acknowledge events, so nothing is missed and every response is recorded. A notification badge in the header shows unacknowledged alerts, and each event can be acknowledged by a named user; acknowledgements, and the full event history, are retained in the database and exportable to CSV (Section 8.3).

- **Filter and review.** Events can be filtered by severity (alert, warning, information, cleared), by acknowledgement state, and by location and date range.

- **Every event type.** The log covers rainfall, flood and pump events, temperature and wind, camera-verification prompts, sensor faults and data-quality alerts, and power/battery status.
- **Acknowledge and audit.** Each event shows who acknowledged it and when (or that it auto-cleared), giving a clear operational audit trail.

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Map Trends Flood Alerts

Alerts & notifications

Full history of every event and alert the system has generated. Filter, review and acknowledge.

3 Active alerts

5 Unacknowledged

28 Today

214 Last 7 days

All Alert Warning Info Cleared Unacknowledged

All locations Last 7 days

Event log

214 events - newest first

Mark all as read

	High-level float WET confirmed - pumps running (Duty 1 & 2).	Marrickville - Flood	18 Jul 14:31	Acknowledge
	Water level +120 mm exceeded pump-start +100 mm - DUTY pumps started.	Marrickville - Flood	18 Jul 14:30	ack P.Nair 14:32
	Rainfall 26.4 mm/hr exceeded 25 mm/hr threshold - flood watch raised.	Marrickville - Rainfall	18 Jul 14:22	Acknowledge
	Rainfall 24.1 mm/hr approaching threshold (>50% of warning).	Belmore triangle - Rainfall	18 Jul 14:20	ack J.Okoro 14:24
	Standing water detected - PTZ verification requested.	Canterbury - Camera	18 Jul 14:10	Acknowledge
	Wind gust 63 km/h - approaching 75 km/h threshold.	Belmore triangle - Wind	18 Jul 13:58	ack J.Okoro 14:01
	Temperature fell below 38 C - heat TSR alert cleared.	Windsor Road SSC - Temperature	18 Jul 13:40	auto-cleared
	Radar sensor comms timeout - fallback to float; data-quality alert.	Campsie - Fault	18 Jul 11:05	ack T.Reilly 11:20
	Battery on solar charge - state of charge 96%.	Marrickville - Power	18 Jul 09:12	auto-cleared
	Up-tunnel level rising - rate-of-rise watch.	Lady Game Drive - Flood	17 Jul 22:44	ack S.Chen 22:50
	Water level receded below +40 mm - pumps stopped (min run met).	Marrickville - Flood	17 Jul 20:15	auto-cleared

Showing 1-11 of 214 - all events are retained in the database and exportable to CSV (Section 8.3).

[Load more](#)

Figure 15 - Alerts and notifications panel: a filterable history of every event and alert (severity, station, category, message, timestamp and acknowledgement status), with summary counts and per-event acknowledge actions; all events are retained and exportable to CSV (demonstration data).

8. Online portal, data transmission and power

8.1 Central server and online portal

The central server visualises every station over a map, monitors the network, generates screen, SMS and email alerts, and stores all history for retrieval and reporting.

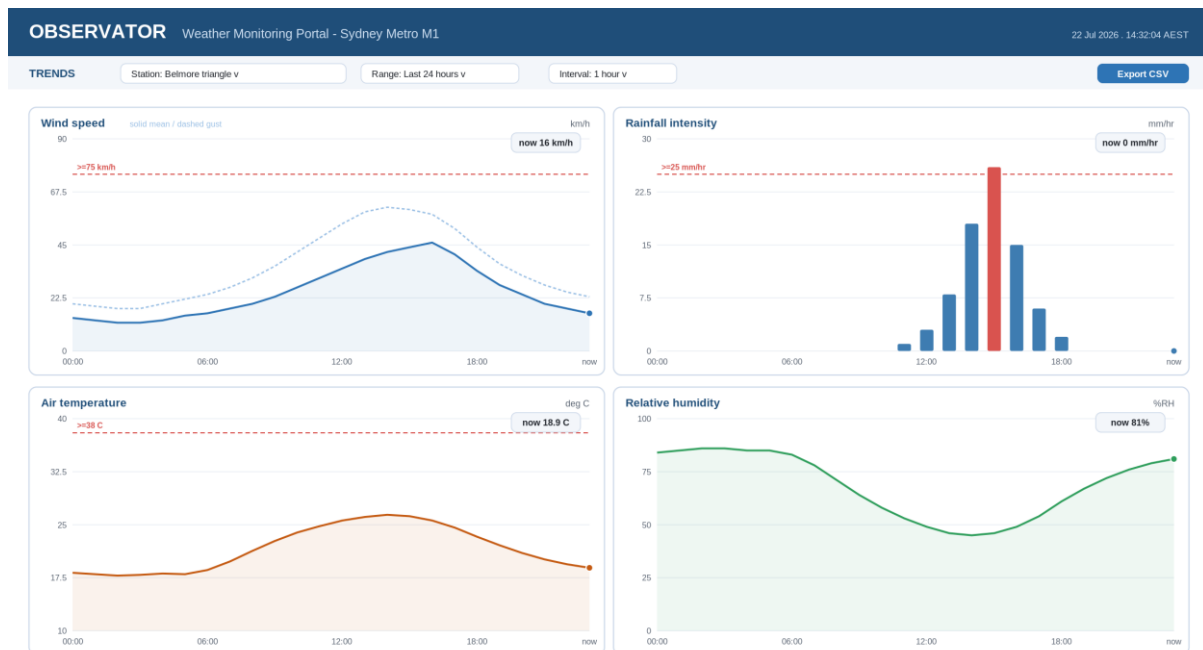
Hosted on the Amazon (AWS) or Google cloud - high availability with no hardware to buy. The software is hosted on the **Amazon Web Services (AWS) or Google cloud** rather than on a physical server that MTS or Observator would have to purchase, house and maintain. This means the **hardware high availability is provided by the hosting company as part of the service**: the underlying compute, storage and networking run across redundant, enterprise-grade data-centre infrastructure with automatic failover, so **MTS does not need to invest in server hardware, spare parts, data-centre space, power or cooling**, and there is no ageing hardware to refresh over the life of the contract. Capacity can be scaled up or down on demand, data is stored durably with automated backups, and the platform benefits from AWS's physical security and resilience. The specific AWS region, availability zones and any data-residency or IT/cyber requirements will be confirmed with MTS during detailed design (see Section 15).

Requirement	Solution response
Displays all specified metrics; always available (48 h notice of maintenance)	Comply. Real-time and historical metrics per parameter and location, on a map view; maintenance notified 48 h in advance.
20 MTS user logins	Comply. 20 named logins (or more) with role-based access.
Stable and uninterrupted during revenue hours	Comply. Hosted on the cloud servers with hardware high availability and automatic failover provided as part of the service (no MTS hardware investment); stability targeted to revenue operating hours.
Optimised HMI for laptop and mobile; intuitive	Comply. Single responsive interface (device-width viewport, fluid grid) that auto-resizes from phone to tablet to laptop/desktop, with touch-friendly navigation and scrollable tables/charts on small screens - one codebase, no separate app (see Figure 9).
Interrogate and update weather-event alerts	Comply. Users can filter, interrogate, annotate and acknowledge alerts.
Display system health / fault indicators, refreshed every 10 minutes	Comply. Live health dashboard refreshing at ≤ 10 -minute intervals.
Historical data storage and retrieval	Comply. Full historical archive with query and export.



Illustrative OMC-data portal view - demonstration data - wind shown in km/h against the ≥ 75 km/h alert threshold

Figure 6 — Illustrative OMC-data portal view: corridor map with each station's readings shown against its alert thresholds, plus system-health fields and a station-status table (demonstration data).



Illustrative OMC-data portal - Trends view - demonstration data (Belmore triangle, last 24 h) - thresholds shown dashed

Figure 7 — Portal Trends screen: time-series charts of wind speed (mean and gust), rainfall intensity, air temperature and relative humidity for a selected station, with alert thresholds shown dashed (demonstration data).

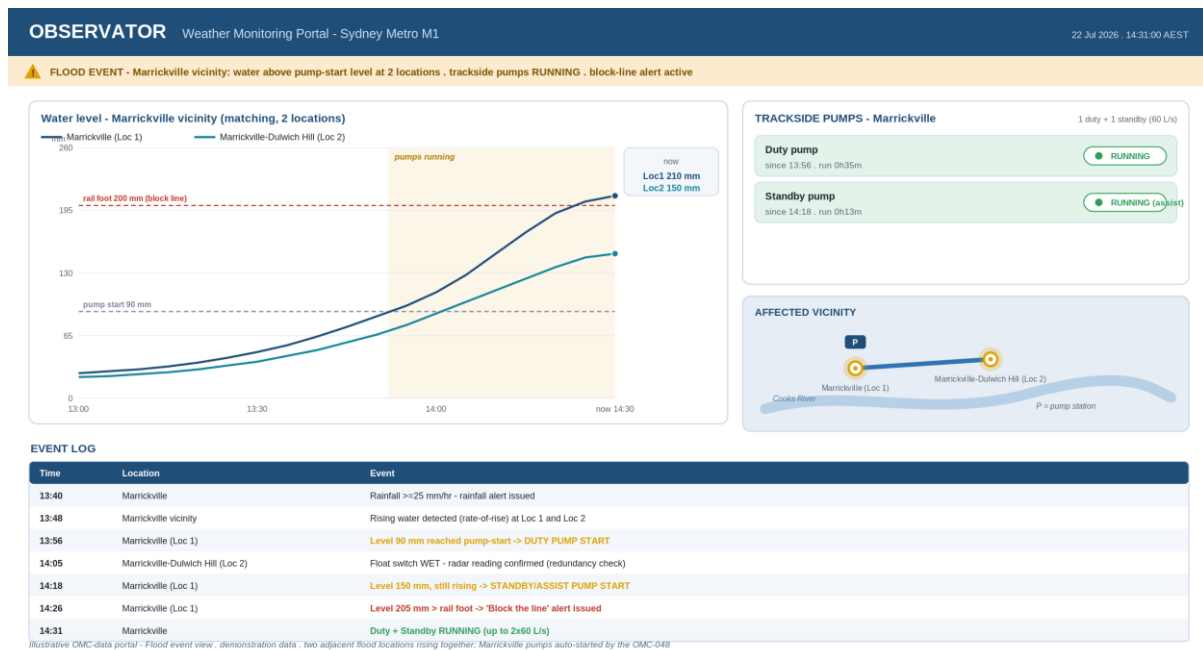


Figure 8 — Portal Flood-event screen: matching water-level time series for two adjacent Marrickville-vicinity locations, with the trackside pumps auto-started (lead duty then standby/assist) as levels cross the pump-start and high-high thresholds, pump status, an affected-vicinity inset and a timestamped event log (demonstration data).

Responsive HMI — one portal, every device

The same server portal auto-resizes: a single column on a phone, two columns on a tablet, and 3–4 columns on a laptop/desktop.

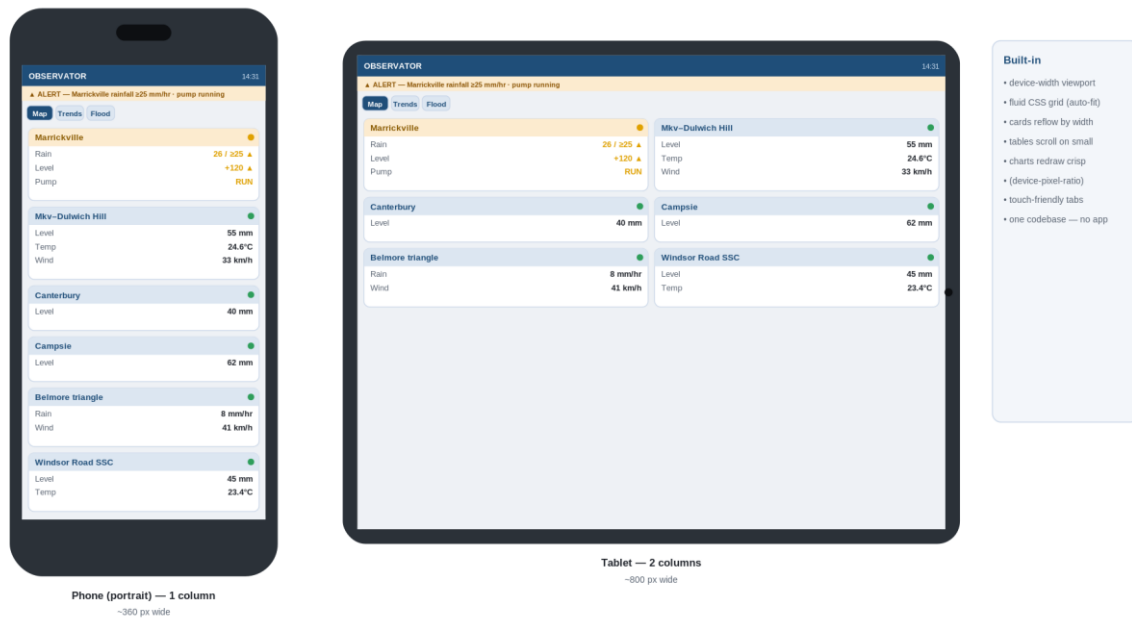


Figure 9 — Responsive HMI: the OMC-data portal reflows automatically from a single column on a phone to two columns on a tablet and 3–4 columns on a laptop/desktop (device-width viewport, fluid grid; demonstration data).

The station and pump-control screens are responsive in the same way. On a phone the Marrickville pump screen reflows into a single, scrollable, touch-friendly column so field and on-call staff can review the sensors and, where authorised, operate the pumps from a mobile device.

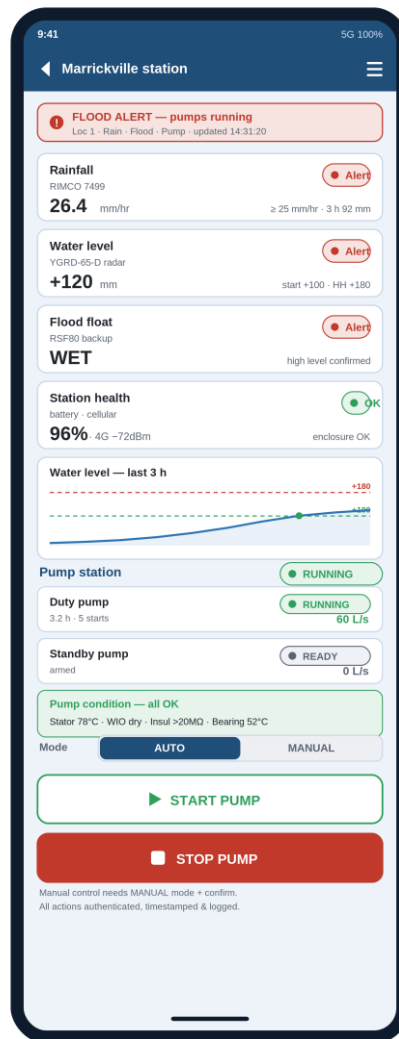


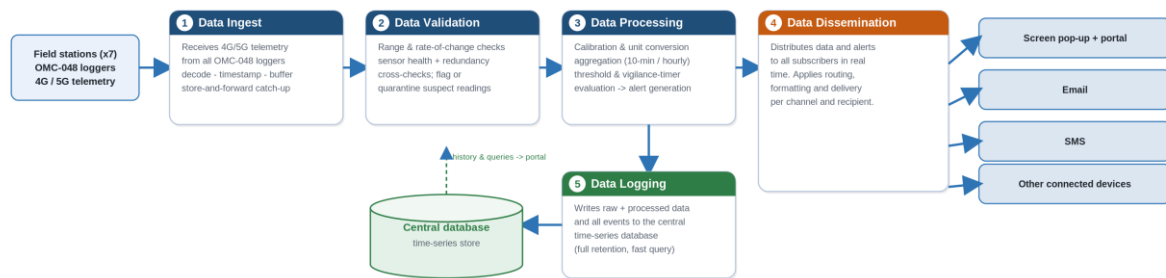
Figure 13 - Mobile view of the Marrickville pump-station screen: the same live sensor readings, water-level trend, pump status and existing-pump condition monitoring, with the AUTO/MANUAL toggle and supervised START/STOP pump control, reflowed into a single touch-friendly column for phones (demonstration data).

8.2 Software architecture and modules

Our software is built as a set of cooperating modules, each with a single clear responsibility. Data flows through the modules in sequence - ingest, validation, processing, logging and dissemination - which makes the system robust, testable and straightforward to extend. The same pipeline runs on the central server and, in a compact form, on each OMC-048 data logger, so that safety-critical detection, alerting and pump control continue locally even if the central link is lost.

Central software — modular data pipeline

Five cooperating modules take field telemetry through to validated, processed data that is logged and disseminated to all subscribers.



All five modules run on the central server; the OMC-048 also performs local ingest, validation, processing and alerting at each station so safety-critical detection and pump control continue independently of the link.

Figure 11 - Central software architecture: the five modules (Data Ingest, Data Validation, Data Processing, Data Logging and Data Dissemination), the central time-series database, and dissemination to screens, SMS, email and other connected devices.

a) Data Ingest

The ingest module receives telemetry from every OMC-048 logger over the 4G/5G network, decodes each message, applies an authoritative timestamp and buffers it. Both routine readings and event-triggered messages are accepted, and the module reconciles any store-and-forward backlog after a communications outage so that no data is lost.

b) Data Validation

Incoming data is quality-checked before it is trusted. The validation module applies range and rate-of-change limits, checks sensor health and status flags, and cross-checks redundant sources (for example radar against float, or a reading against a neighbouring station). Readings that fail are flagged or quarantined and raise a data-quality alert, so operators are never shown misleading values and designated fall-back sources can take over automatically.

c) Data Processing

Validated data is turned into meaningful information. The processing module applies calibration and unit conversion, derives aggregates (such as 10-minute, hourly and multi-day rainfall totals), and evaluates the configured thresholds, dwell timers and vigilance/reset logic. When a threshold is breached it generates the corresponding alert with the correct wording and severity for downstream distribution.

d) Data Dissemination

The dissemination module is responsible for distributing the data to its subscribers. It applies per-channel routing, formatting and delivery so the same event reaches the right people and systems in the right form: on-screen pop-ups and the live portal, SMS, email, and other connected devices or systems (for example operational displays or third-party interfaces). Delivery is confirmed and retried on failure, and recipient lists and routing rules are fully configurable.

e) Data Logging

Every raw reading, processed value and event is written to the central time-series database, giving a complete and permanent record for the whole network. The logging module underpins the historical query engine and CSV export described in Section 8.3, and provides the audit trail for incident review and reporting.

8.3 Historical data, query engine and CSV export

In our software, every reading from every sensor is written to a **central database** on the server, giving a complete, permanent history for the whole network. Historical data can be easily retrieved using a **simple-to-**

use query engine: the operator specifies the search parameters - **by location, by sensor/parameter and by date range** (with an optional averaging interval) - and runs the query without any technical knowledge or SQL.

The matching records are **displayed on screen** in a sortable, paged results table, and the exact filtered result set can be **easily exported in CSV format** with a single click for use in Excel or any external analysis, reporting or archiving tool. This supports incident review, trend analysis, regulatory reporting and data hand-over to MTS.

- **Stored in a database.** All sensor readings are retained in the central time-series database with full history and fast retrieval.
- **Simple query engine.** Search by location, sensor/parameter and date range (plus optional interval) - point-and-click, no technical skills required.
- **On-screen results and CSV export.** Results display in a paged, sortable table and export to CSV in one click for external analysis and reporting.

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Map Trends Flood History

Historical data — query engine

All sensor readings are stored in the central database. Retrieve any history by location, sensor and date range; view on screen and export to CSV.

Location(s): Marrickville, Belmore triangle | Sensor / parameter: Rainfall, Water level | From date: 01 Jul 2026 | To date: 18 Jul 2026 | Interval: 10-minute | Run query | Export CSV

Query results 1,248 records found · showing 1–14 · sorted by time (newest first) SELECT * WHERE loc IN (...) AND param IN (...) AND ts BETWEEN 2026-07-01 AND 2026-07-18

Timestamp	Location	Parameter	Value	Unit	Status	Logger / ID
2026-07-18 14:30	Marrickville	Rainfall	26.4	mm/hr	Alert	MKV-RIMCO-01
2026-07-18 14:30	Marrickville	Water level	+120	mm	Alert	MKV-YGRD-01
2026-07-18 14:20	Marrickville	Rainfall	24.1	mm/hr	Warning	MKV-RIMCO-01
2026-07-18 14:20	Belmore triangle	Water level	72	mm	Normal	BEL-YGRD-01
2026-07-18 14:10	Belmore triangle	Rainfall	9.6	mm/hr	Normal	BEL-RIMCO-01
2026-07-18 14:00	Marrickville	Water level	+95	mm	Warning	MKV-YGRD-01
2026-07-18 13:50	Belmore triangle	Water level	61	mm	Normal	BEL-YGRD-01
2026-07-18 13:40	Marrickville	Rainfall	18.7	mm/hr	Normal	MKV-RIMCO-01
2026-07-18 13:30	Belmore triangle	Rainfall	6.2	mm/hr	Normal	BEL-RIMCO-01
2026-07-18 13:20	Marrickville	Water level	+70	mm	Normal	MKV-YGRD-01
2026-07-18 13:10	Belmore triangle	Water level	55	mm	Normal	BEL-YGRD-01
2026-07-18 13:00	Marrickville	Rainfall	12.9	mm/hr	Normal	MKV-RIMCO-01
2026-07-18 12:50	Belmore triangle	Rainfall	4.1	mm/hr	Normal	BEL-RIMCO-01
2026-07-18 12:40	Marrickville	Water level	+58	mm	Normal	MKV-YGRD-01

Rows per page: 25

Data source: central time-series database (all stations, full retention). Export delivers the exact filtered result set as a UTF-8 CSV file.

Figure 10 — Historical-data query engine: filter by location, sensor and date range; matching records shown on screen in a paged table with per-reading status, and one-click Export to CSV (demonstration data).

8.4 User management and access control

The portal includes built-in user management, so MTS retains full control of who can access the system and what they can do, with no dependence on Observator for day-to-day account changes. A designated MTS administrator manages all accounts from an administration console; every user has a named, individually authenticated account, and access to features and data is governed by role-based access control (RBAC) on a least-privilege basis.

Adding and removing users

The administrator can add, edit, temporarily suspend and permanently remove users at any time, so access always matches MTS current staff and duties.

- **Add a user.** Create an account with the person name, email and mobile number (for SMS alerts), assign one or more roles, and choose which stations or areas they may access. New users receive a secure invitation to set their own password.

- **Remove or suspend a user.** Access can be revoked instantly when someone leaves or changes role - an account can be suspended (blocking login while retaining its audit history) or permanently deleted.
- **Edit a user.** Roles, station scope, contact details and notification preferences can be changed at any time and take effect immediately.
- **Scale.** The system supports the number of concurrent MTS logins required (20 or more) with no functional limit beyond the licensed user count.
- **Multi-factor authentication (optional).** Multi-factor authentication (MFA) can be enabled for portal access as an additional security control.

Roles and permissions

Rather than granting permissions one by one, the administrator assigns each user one or more roles; each role bundles a defined set of permissions following least privilege. A set of standard roles is provided and can be tailored, and additional custom roles can be created to match MTS operating model. A user may hold more than one role, and each user access can be scoped to specific stations or areas so people see only what is relevant to them. The standard roles are summarised below.

Role	Typical user	Key permissions
Administrator	System owner / IT	Full control: manage users and roles, configure thresholds and alert recipients, and all view and control functions.
Operator	OCC / control room	View all live dashboards and history, receive and acknowledge alerts, and initiate operational responses; cannot manage users or change configuration.
Pump Controller	Authorised operations staff	All Operator permissions plus supervised manual pump start/stop (MANUAL mode with confirmation) at permitted stations.
Maintainer / Technician	Field and maintenance	Device and power health, calibration and fault flags, and maintenance mode; acknowledge maintenance alerts; limited configuration.
Analyst / Reporting	Engineering and reporting	Historical query engine, trend analysis and CSV export (Section 8.3); no live control.
Viewer (read-only)	Stakeholders	View dashboards and history only; no actions or changes.

Accountability, audit and security

Because every account is named and individually authenticated, every action is attributable to a specific person.

- **Named accountability.** No shared logins; every login and significant action is tied to a named user.
- **Full audit trail.** User additions and removals, role changes, threshold and configuration edits, alert acknowledgements and manual pump start/stop are recorded with user, timestamp and detail, and are queryable and exportable.
- **Separation of duties and least privilege.** Sensitive actions such as manual pump control are restricted to authorised roles and require MANUAL mode plus confirmation.
- **Session and password security.** Enforced password policy and account lockout, encrypted sessions, automatic inactivity timeout, and a permission re-check on every action.

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Map Trends Flood History **Admin**

User management

Add, edit, suspend or remove users and assign their roles and station access. Administrator only.

Q Search name, email or role Role: All 8 users · 7 active · 1 suspended + Add user

User	Role(s)	Station access	Status	Last login	Actions
SC Sarah Chen s.chen@metrotrains.com.au	Administrator	All stations	Active	Today 14:22	Edit Remove
JO James Okoro j.okoro@metrotrains.com.au	Operator	All stations	Active	Today 14:05	Edit Remove
PN Priya Nair p.nair@metrotrains.com.au	Pump Controller Operator	Marrickville	Active	Today 13:50	Edit Remove
TR Tom Reilly t.reilly@contractor.com	Maintainer	All stations	Active	Yest 16:10	Edit Remove
AK Aisha Khan a.khan@metrotrains.com.au	Analyst	All stations	Active	2 days ago	Edit Remove
DS David Smith d.smith@metrotrains.com.au	Viewer	Belmore, Campsie	Active	3 days ago	Edit Remove
ML Mark Lee m.lee@metrotrains.com.au	Operator	All stations	Suspended	12 Jul	Edit Remove
EW Emma Wilson e.wilson@metrotrains.com.au	Viewer	All stations	Invited	—	Edit Remove

Roles (least-privilege; a user may hold more than one; access can be scoped to selected stations)

- Administrator** — full control incl. user management & configuration
- Pump Controller** — Operator + manual pump start/stop (MANUAL + confirm)
- Analyst** — historical query, trends & CSV export
- Operator** — live dashboards, acknowledge & action alerts
- Maintainer** — device health, calibration & maintenance mode
- Viewer** — read-only dashboards & history

Figure 14 - User management console: the administrator adds, edits, suspends or removes users, assigns one or more roles and scopes each user station access; the panel lists each user with role chips, status and last login, with role definitions shown below (demonstration data).

8.5 Data transmission

- End-to-end communications from every sensor to the central server are provided by Observator, using **4G/5G cellular** as preferred, with store-and-forward buffering on the OMC-048 logger during any outage.
- Electromagnetic compatibility:** equipment is selected and installed so as not to interfere with rolling-stock systems or infrastructure during transmission; EMC compliance evidence provided during design.

8.6 Autonomous power

As the Statement of Requirements assumes no power source at the locations, each station is powered by an autonomous solar power system supplied by Observator. It comprises a **400 W solar panel**, **two 55 Ah LiFePO4 (lithium iron phosphate) batteries (2 x 55 Ah)** and a **solar charge controller**, sized to carry the station load with multi-day autonomy through overcast weather. The existing Marrickville pumps use their existing electrical supply. The **solar charge controller reports the battery health and charge status - state of charge, PV input, charge/discharge and fault flags - to the OMC-048 data logger, and this information is displayed live in the portal** and alarmed as a leading indicator of failure, so power-system problems are visible and actionable before they cause an outage.

9. System performance and availability

Requirement	Observer response
Planned-maintenance downtime — one 36 h period/year aligned to weekend possessions	Comply. Planned maintenance and calibration scheduled into a single 36-hour window per year, coincident with planned weekend possessions.
Availability — not inoperable >12 h in any 6 months (operating hours) and ≤24 h cumulative per 12 months	Comply. Redundancy, remote diagnostics and local spares hold outages within these limits.
System status — report inoperable / unresponsive / missing sensor or pump data / leading indicators (e.g. low battery)	Comply. Continuous self-diagnostics report each condition and raise system-fault alerts.
Redundancy — alerts from alternate source on failure, with reworded coverage	Comply. Automatic fallback per Section 7.4.
Re-calibration — advise when required (e.g. after CM); provide calibration certificates	Comply. Blue2Care advises recalibration needs and supplies calibration certificates once systems are calibrated.
Configuration management — CCB approval for non-like-for-like and portal changes	Comply. Managed under the Principal's Configuration Management Plan; see Section 13.

10. Implementation, installation and site works

10.1 Implementation plan and programme

Observer delivers the works as a staged, stage-gate programme under a single project manager who is accountable to MTS for the full scope. The indicative programme spans about 20 weeks from contract award to practical completion, followed by ongoing warranty and support. Off-corridor work - build, configuration, factory testing and server staging - is front-loaded to retire risk before any track access is used, and the on-corridor installation is planned around confirmed possession windows so each access event delivers maximum progress. Durations are indicative and confirmed with MTS at initiation; the programme is driven principally by the availability of rail access windows.

Stage	Weeks	Key activities and deliverables
1 Initiation and mobilisation	W1-2	Kick-off; project management, WHS and quality plans; access/possession strategy; baseline programme and risk register.
2 Detailed design and site surveys	W2-6	Site surveys at all seven locations; detailed design; threshold and alert configuration; design approval (hold point).
3 Procurement, build and FAT	W5-12	Procure long-lead items (loggers, sensors, pump set); assemble and wire enclosures; configure loggers; factory acceptance test per station.
4 Server build and staging	W8-12	Stand up the central server, portal and alert engine; prove station-to-server comms in staging.
5 Site installation	W12-18	Install masts, enclosures, sensors, solar/battery and (at Marrickville) the pump set; each site made safe each shift.
6 Commissioning and SAT	W15-19	Power-up, integration and a per-location site acceptance test against the server; ITP sign-off with MTS.
7 Handover and training	W19-20	As-builds, O&M manuals and certificates; operator and maintenance training; practical completion.
8 Warranty and support	Ongoing	Remote monitoring, preventive maintenance and defect rectification under Blue2Care (Section 11).



Figure 16 - Implementation programme (weeks from contract award): off-corridor stages (build, configure, FAT, server staging) run in parallel and complete before minimal on-corridor works (install, SAT); key milestones are design approval, FAT and SAT sign-off, and practical completion.

- **Key milestones.** Design approval (end Week 6); FAT sign-off for all stations (end Week 12); SAT sign-off and system live (end Week 19); practical completion and handover (Week 20).
- **Ensuring timely delivery.** Front-loading off-corridor work retires most technical risk before site; the server build runs in parallel with procurement and manufacture; and installation is grouped by locality so a single possession delivers maximum on-site progress. Progress is reported to MTS at an agreed cadence with early-warning of any emerging delay and a recovery action (Section 13).

10.2 Installation and site works

- **Access and possessions.** Works on or near track are performed only during Engineering Hours or agreed weekend possessions, observing the NDI-TAL access windows and the alternate-protection weekend windows in Section 4, and coordinated with other contractors. Restricted-access requests are submitted at least two weeks ahead per the T-minus submission guideline.
- **Siting.** Locations are chosen for accessibility (outside transit space, beyond overhead-wire safe approach distances), security (inaccessible to the public), relevance (flood areas of interest) and clearance from obstructions for wind monitoring.
- **Height access.** Observer provides height-access equipment operated by licensed personnel, with the height-access schedule approved in writing by the Principal before work.
- **Competency.** All personnel hold the required RIW cards, MTS inductions, Electrical Safety Rules awareness and site-specific training before commencing; organisation and site-management charts are provided per Section 6 of the Statement of Requirements.
- **Storage & housekeeping.** Materials are stored only with written approval; no dangerous goods on Principal premises; work areas are left clean and surplus materials removed after each activity.

11. Maintenance and support (Blue2Care)

Preventative and corrective maintenance, commissioning, calibration and emergency management are delivered under Observer's Blue2Care program, resourced from the Australian office.

11.1 Preventative maintenance

- Delivered to Observer's Technical Maintenance Plans, meeting statutory obligations, OEM specifications and industry standards, without relying on latitude.
- Performed during Engineering Hours or possessions on wayside equipment, with a high standard of care to minimise corrective work.
- All measuring instruments are calibrated by recognised laboratories on a regular basis, labelled, and listed with calibration certificates submitted annually to the Principal for approval.

11.2 Response times

Service request	Required
Corrective maintenance to rectify a terminal system fault	Response $\leq$ 6 hours — Comply.
On-site investigation and proposed remediation of system failure	Within 12 hours — Comply.
On-site repair services including fault remediation	Within 24 hours — Comply.
Maximum time to repair — system inoperable (from start of remediation)	Within 24 hours — Comply.

12. Safety, environment and compliance

- **Work Health and Safety.** Compliance with WHS legislation; safe work method statements and task risk assessments; notifiable incidents actioned within 24 hours; written reports on safety incidents and observations.
- **Rail Safety National Law.** Compliance with RSNL duties; safe railway operations agreed with the Principal, OpCo and Sydney Metro; a drug and alcohol management program; competency records aligned to the AQF; cooperation with the Office of the National Rail Safety Regulator.
- **Protection of the Environment Operations Act.** Environmental management systems; pollution incidents reported to the Principal (and, for major incidents, to 000); records maintained for compliance.
- **Modern Slavery Act.** Compliance with applicable Modern Slavery laws; annual statutory declaration; supply-chain due diligence; cooperation with the Principal's audit rights.
- **Security.** Physical security measures per rail-industry, SMA, ARTC, Sydney Trains and Principal standards; anti-vandal mounting given expected vandalism, theft and graffiti along the Southwest corridor; prompt reporting of breaches.
- **Information/OT security.** Compliance with the OT Supplier Security Management Standard (SMCSWTS2-MTS-1NL-IT-PRO-005862); prevention of unauthorised access to Principal or third-party equipment and data.

13. Governance and contract management

- **Representatives.** A Contract Management Representative is nominated within 15 days of the Agreement as single point of contact, with a Project Manager for day-to-day activities; organisation and site-management charts are provided one day after the Start Date.
- **Meetings.** Attendance at safety tool-box, fortnightly, monthly performance-review and any emergency meetings, with delegated authority to act.
- **Incident management.** Allocated incidents responded to within the agreed timeframes, with analysis, impact assessment and cooperation with other contractors.
- **Variations.** Additional services quoted against the Schedule of Rates, with written variation proposals within five business days where required.
- **Configuration management.** Changes managed under the Principal's Configuration Management Plan; like-for-like replacements treated as maintenance, non-like-for-like changes submitted to the CCB.
- **Audit and reporting.** Support for random audits at no additional cost, with corrective actions implemented and closed; performance reporting per agreed schedules.
- **Invoicing.** Itemised monthly invoices.

13.1 Contract plans

A Safe Work Method Statement will be submitted within 15 days before site works, with each plan kept to no more than five pages specific to scope, as required by Section 17.2 of the Statement of Requirements.

14. Summary of compliance

Observator confirms that the proposed solution addresses the full scope of the Statement of Requirements: rain, flood-level, temperature and wind monitoring at the seven nominated locations; two automatically triggered trackside flood pumps at Marrickville; data transmission, processing and configurable multi-stage alerting; a secure 20-user online portal; and preventative and corrective maintenance with calibration — all delivered by a single ISO 9001:2015-certified vendor with local Australian support. Items requiring confirmation are limited to those listed in Section 15 and do not affect the overall approach.

15. Assumptions, exclusions and clarifications

In the interest of a transparent response, Observator notes the following. None is considered a barrier to delivery; each will be closed out during site visits and detailed design.

Confirmations sought from MTS

- Final kilometrages and GPS positions for all seven locations (to be confirmed on site visits).
- Confirmation of all threshold values and alert wording currently marked "pending confirmation" in the Statement of Requirements (rainfall, flood, temperature and wind).
- Confirmation of the nominated MTS alert recipients (email and mobile) and portal user list (20 logins).
- Availability and rating of the 240 V GPO supply and the existing PVC outfall at Marrickville.
- Access to Annexure A track diagrams and Annexure B technical documentation to finalise mounting and interface details.
- PTZ camera live-feed web (URL) links for the relevant locations, plus any authentication and network access needed to reach them, so the links can be embedded in the screen, email and SMS alerts (Section 7.5).

Technical items to be verified against current datasheets / calibration

- **Rainfall accuracy.** The RIMCO 7499 BOM-standard gauge is the rainfall compliance instrument; its $\pm 2\%$ (≤ 50 mm/hr) tolerance is confirmed by factory calibration certificate. Rainfall is monitored only at the locations that require it.
- **Wind range - resolved.** The Gill WindSonic 75 measures the full 0-75 m/s (270 km/h) range required by the Statement of Requirements, so no substitution is needed.
- **Temperature and humidity - resolved.** The Gill GMX300 compact sensor measures air temperature and relative humidity (and barometric pressure) mounted at 1.5-2 m, satisfying the radiation-screen and siting requirements independently of the 10 m wind mast. Its integrated radiation screen and ± 0.3 °C accuracy meet the certified-screen requirement without a separate shield.
- **Other specs.** Confirmation of IP ratings and the +70 °C temperature ceiling on the selected temperature sensor (radar level accuracy is datasheet-confirmed at ± 1 mm).

Commercial / supply notes

- Sensors (Gill WindSonic 75, Gill GMX300 temperature/humidity sensor, RIMCO 7499 rain gauge, YGRD-65-D radar level sensor and RS PRO RSF80 float switch), the OMC-048 data logger and the central server are supplied and integrated by Observator as a single managed system.
- The trackside flood pumps already exist on site; Observator integrates to the existing pumps and their control gear rather than supplying pumps. The OMC-048 logger provides automatic real-time control (start/stop demand to the existing pump control panel) and reads pump run/fault status for alerting. The interface points on the existing panel, and the existing pumps' electrical supply, are confirmed at detailed design.
- The specified YGRD-65-D radar level sensor is a third-party instrument; Observator will confirm its rail EMC compliance (no interference with rolling stock, per Section 8.5) and local support and spares arrangements during detailed design.
- This document is a technical response to Part A (Statement of Requirements); commercial pricing, the Schedule of Rates and contract conditions are provided separately.

Appendix A - Existing trackside pump integration (Marrickville)

The trackside flood pumps at Marrickville already exist on site. Observator does not supply pumps; the OMC-048 data logger integrates to the existing pumps and their control gear, providing automatic real-time start/stop control and reading pump status for alerting and reporting. The signals below are the typical interface points; the exact terminals, ratings and available status signals on the existing pump control panel are confirmed during detailed design (Section 15).

Integration interface (typical)

Signal	Direction / description
Start/stop demand - duty (lead)	OMC-048 volt-free relay output to the existing pump control-panel remote-start / contactor input.
Start/stop demand - standby / assist	Second OMC-048 volt-free relay output; energised at the high-high level or on a duty-pump fault.
Pump run confirmation	Digital input to the OMC-048 from the existing panel's run / status auxiliary contacts, per pump.
Pump fault / trip	Digital input to the OMC-048 from the existing panel's common-fault / motor-protection auxiliary contacts.
Pump condition (where available)	Any existing condition-monitoring outputs (for example a fitted Grundfos IO 113, or temperature / seal alarms) are read into the OMC-048 for display and alerting.
Level demand source	The OMC-048 derives the start/stop demand from the YGRD-65-D radar level (primary) and the RSF80 float switch (independent backup).

Notes and assumptions

- Observator supplies the interface wiring and any signal-conditioning / relay interface required between the OMC-048 and the existing pump control panel; the pumps, motors, pipework, outfall and existing control panel remain the existing site assets.
- The existing pump control panel is assumed to accept a remote / automatic start-stop demand and to provide run and fault status. Where it does not, a small interface kit or minor panel modification (by a licensed electrician) is scoped with MTS.
- The existing pumps' electrical supply, ratings, duty and outfall are as installed; no change to the pump hardware is proposed. The duty/standby control logic and thresholds are implemented in the OMC-048 (Section 5.6) and confirmed with MTS.